

Bayesian VAR and TVP-VAR Models: Do Different
Demographics and Socio-Economic Groups
Experience Monetary Policy Differently?

by

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Abstract

VAR and TVP-VAR models have been a long-standing staple of the economic literature for forecasting and impulse response analysis, yet have seen no use for interpreting the impact of monetary policy on different socio-economic and demographic groups. We use a dataset of 24 economic variables, of which 7 are related to dis-aggregated socio-demographic data, to fit a variety of constant coefficient and time-varying parameter VAR models. Different regularization and stochastic volatility specifications are implemented to help deal with multicollinear unemployment data and over-parameterization concerns. We compare and contrast these models and find that the horseshoe and normal-gamma priors display the best forecasting performance for constant coefficient and time-varying parameter models, respectively. Finally, we analyze the coefficients of the best fitting models and recover results from empirical economic theory, a relationship between the Federal Funds rate and the wealth of the bottom 50% of the United States population and possible evidence of omitted variable bias.

Contents

1	Introduction	4
1.1	Vector Autoregressive Models	5
1.2	Time-Varying Parameter Models	6
1.2.1	Time-Varying Parameter VAR Model	6
1.2.2	Time-Varying Parameter Regressions	7
1.3	Introducing Stochastic Volatility	8
1.3.1	Cholesky Decomposition	8
1.3.2	Factor Stochastic Volatility	9
2	Estimation and Regularization	11
2.1	Bayesian vs. Frequentist Approach	11
2.2	Model Evaluation and Interpretation	12
2.2.1	Evaluation of Point Forecasts	12
2.2.2	Predictive Density Scores	12
2.2.3	Impulse Response Functions	13
2.3	Choice of Prior and Regularization Priors	14
2.3.1	Bayesian Lasso	14
2.3.2	Stochastic Search Variable Selection	15
2.3.3	Global-Local Shrinkage	16
2.3.4	Horseshoe Prior	16
2.3.5	Normal-Gamma Prior	16
2.3.6	Triple-Gamma Prior	17
3	Data and Methodology	18
3.1	Exploring the Data	18
3.2	Gibbs Sampling for TVP-SV-VARs	20
3.3	Models To Be Fit	20

4	Results and Discussion	22
4.1	Forecasting Errors	22
4.2	Log Predictive Density Scores	24
4.3	Factor Loadings	24
4.4	Impulse Response Functions	27
4.5	Structural Matrix for the TVP-SV-VAR-NG	28
5	Conclusions	31
A	Some Conditional Posterior Distributions	32
B	Additional Information for the Triple-Gamma Prior	36
C	Code for Computing RMSFE and Joint RMSFE For Time-Varying Parameter Models	38
D	Posterior Median Point Estimates for RMSFE and Joint RMSFE and State Plots	40
E	Supplementary Resources	45

Chapter 1

Introduction

The role of central banks has evolved significantly over the last several decades. Prior to the 1970s, central banks traditionally focused on monetary stability. Beginning in the early 1990s, many central banks around the world began explicitly targeting a 2-3% inflation and a low unemployment rate. This dual mandate is inspired by the belief that an economic environment with both low inflation and low unemployment will promote long term and sustainable economic growth. Achieving this goal requires an understanding of inflation’s relationship with other macroeconomic variables, global events, and other factors. Central bankers must ensure the consequences of monetary policy changes are foreseeable. A framework that can simulate and forecast these changes in inflation and unemployment can be an extremely valuable policy and decision-making tool. Accurate forecasting becomes increasingly important, as maintaining economic stability and growth is a delicate process that requires careful consideration.

Vector Auto-regressive (VAR) models have been a groundbreaking model architecture in the econometric literature since their introduction in [Sims \(1980\)](#), and for good reason. Before the introduction of VAR models, inflation forecasting literature typically relied on large-scale simultaneous equation models that imposed arbitrary restrictions to guarantee that the model parameters could be uniquely estimated from the data, ensuring identifiability. VAR models were proposed as an alternative, providing a flexible framework to simultaneously analyze complex inter-dependencies between key economic variables without arbitrary restrictions. Academia, policy-making institutions and central banks have used VAR models to determine the effects of monetary policy, economic and financial shocks, and business cycle fluctuations.

While VAR models have been extensively used to study the effects of monetary policy, most applications have focused on forecasting, overlooking this model’s ability to interpret relationships across demographic variables. In this paper, we hope to address this discrepancy by using time-varying parameter VAR models with appropriate regularization to determine whether or not different groups experience these policy and macroeconomic shocks differently. Specifically, we will incorporate data on unemployment rates by ethnicity and gender, as well as measures of wealth distribution. By analyzing impulse responses and fitted model coefficients, we can interpret and evaluate the existence of differing relationships with monetary policy across socio-demographic and economic groups. These results might have important implications for policymakers, as it can help them better understand the consequences of monetary policy across their population, and guide their decisions in a way that may be more effective for their goals.

1.1 Vector Autoregressive Models

Uni-variate auto-regressive (AR) models assume that the value of a variable x_t , at time t , depends only on past values (lags) of itself. An $AR(p)$ model allows x_t to depend on its previous p values, which we can formulate as $x_t = c + \sum_{i=1}^p \phi_i x_{t-i} + \varepsilon_t$, where x_t is our target variable at time t , c is a constant, ϕ_i are coefficients, and ε_t is a error term, with mean zero and constant variance. The collection $\{\varepsilon_t\}_{t=1}^N$ of error terms are assumed to be independent and identically distributed, and serially uncorrelated over time, meaning that $cov(\varepsilon_i, \varepsilon_j) = 0 \forall i \neq j$.

VAR models are a multivariate extension of the AR model, designed to capture the dynamic interactions among multiple time series variables. Instead of univariate data, VAR models analyze a vector y_t containing several variables. In a VAR model, current values of each variable in y_t not only depend on their own past values but also on the past values of the other variables. In other words, it allows us to understand how $y_{i,t}$, the i th variable at time t is affected by $y_{j,t-1}$ (or any of the other corresponding lags of variable j). This property allows VAR models to effectively capture complex inter-dependencies, making them especially useful in fields like economics. Understanding the interplay between macroeconomic variables is crucial for policy-making and understanding the impact of policy choices. VAR models provide a robust framework for modeling these relationships, providing a comprehensive analysis of how variables influence one another over time.

A VAR(p) model can be expressed as follows (see [Lubik and Matthes \(2015\)](#); [Koop and Korobilis \(2010\)](#)):

$$y_t = c + \sum_{i=1}^p B_i y_{t-i} + e_t \quad (1.1)$$

Where c is a $k \times 1$ vector of intercepts, B_i are $k \times k$ matrices of lag coefficients for each lag $i = 1, \dots, p$ and the residuals (error terms) $e_t \sim \mathcal{N}(0, \Sigma)$ with $\mathcal{N}$ indicating a multivariate normal distribution with zero-vector mean and Σ a $k \times k$ covariance matrix with $\frac{k(k+1)}{2}$ unique elements. This to fully define the model, we require $(1 + kp)k + \frac{k(k+1)}{2}$ parameters.

The assumptions for the VAR model above are similar to those for linear regression or AR(p) models. We assume a linear relationship between y_t and its lags, and that the data is stationary, with constant mean and variance over time. Another important assumption is homoscedasticity, indicating constant variance of the error terms over time. Additionally, the typical VAR(p) model assumes no multi-collinearity among the explanatory variables, meaning they are not highly correlated to each other. It is also implicitly assumed that the coefficients on the lags (B_i) are constant over time. Thus, we refer to equation 1.1 as a constant coefficient VAR model.

In an economic context, many of these assumptions need to be relaxed. Macroeconomic variables of interest to policymakers and central banks often violate these assumptions. Empirically, variables like inflation, unemployment, and GDP growth exhibit changing volatility over time, challenging the homoscedasticity assumption. For instance, inflation is influenced by a variety of factors such as technological advancements, fiscal policy, central bank policy, and global events. These drivers are constantly evolving, making a model with constant variance inadequate for capturing these dynamics. multi-collinearities are also be hard to avoid in economic data, as important economic indicators like income, inflation and unemployment are inherently correlated.

Other VAR model considerations quickly reveal the need for a different approach. In a

VAR(p) model with k variables, we require $(k + 1) * p + \frac{k(k+1)}{2}$ parameters to fully define the model. As the number of variables k increases, the number of required observations to maintain identifiability grows proportionally to k^2 . This poses a significant challenge when data on macro-economic variables such as GDP, are typically available only on a quarterly basis. This limits the number of observations for each variable, restricting the size of our final dataset. The issue is further compounded if we wish to use a large number of lags p , which increases the risk of overfitting. Overfitting occurs when a model fits the historical data too closely, capturing spurious correlations that do not generalize to new data. To avoid this, we prioritize models that perform well on new observations (out-of-sample) rather than those that only excel on the data used for fitting (in-sample).

To enhance forecasting performance of VAR models, regularization techniques like sparsity and shrinkage are employed. These techniques are designed to prevent a model from becoming too complex by reducing its effective dimensionality. Shrinkage imposes constraints on the size of model parameter values, while sparsity assumes that in a large model, a significant portion of parameters should be exactly zero, with only a subset of variables significantly contributing to the model. For regressions, some of the most commonly used regularization methods include ridge regression and the lasso, both of which can be adapted to a VAR model. By adding an additional penalty terms, we can reduce the size of coefficients to avoid overfitting. The lasso shrinks regression coefficients towards zero using the L1 norm, as follows:

$$\hat{\beta}_{lasso} = \arg \min_{\beta} (||y - \hat{y}||_2^2 + \lambda ||\beta||_1) \quad (1.2)$$

Instead, ridge regression performs the job of shrinkage by using an L2 norm penalty term, which is important for dealing with multi-collinearities. Lasso regularization imposes sparsity and variable selection by setting coefficients exactly to 0. Other forms of regularization will be discussed in a future section.

1.2 Time-Varying Parameter Models

1.2.1 Time-Varying Parameter VAR Model

We introduce an extension of the VAR model where the constant coefficient assumption is relaxed, allowing for time-varying coefficients. The Time-Varying Parameter VAR (TVP-VAR) is defined as follows (Primiceri, 2005; Cogley and Sargent, 2005):

$$\begin{aligned} y_t &= c_t + \sum_{i=1}^p B_{i,t} y_{t-i} + \varepsilon_t, & \varepsilon_t &\sim \mathcal{N}(0, \Sigma) \\ c_t &= c_{t-1} + \eta_{c,t}, & \eta_{c,t} &\sim \mathcal{N}(0, \Sigma_c) \\ \text{vec}(B_{i,t}) &= \text{vec}(B_{i,t-1}) + \eta_{i,t}, & \eta_{i,t} &\sim \mathcal{N}(0, \Sigma_i), \quad i = 1, \dots, p \end{aligned} \quad (1.3)$$

Where c_t is a $k \times 1$ vector of time-varying intercepts, $B_{i,t}$ are $k \times k$ matrices of lag coefficients for each lag $i = 1, \dots, p$ and Σ is a $k \times k$ covariance matrix for the residual errors. Σ_i is a $k^2 \times k^2$ covariance matrix for each matrix of lag coefficients, which many times is assumed to be diagonal, and determines the behavior of the random walk of the time-variation. The $\text{vec}(B_{i,t})$ function, which denotes the vectorization of matrix $B_{i,t}$, indicates that the columns are vertically stacked into a single vector. We additionally assume that ε_t , $\eta_{c,t}$ and $\{\eta_{i,t}\}_{i=1}^p$ are independent.

The number of parameters in the above TVP-VAR model increases (at least) quadratically with the number of variables. Allowing for time-variation in the coefficients introduces

compounded concerns of over-parameterization, as the number of fitted VAR coefficients to estimate grows by a factor of T , the number of observations. This increase exacerbates the risks of overfitting and poor out-of-sample forecasting performance, as discussed for constant coefficient models. Although the introduction of time-variation allows the model to adapt to evolving factors, like changes in monetary policy, it also increases the risk of capturing ephemeral, short-term trends that are present in historical data but unlikely to reoccur. As a result, many papers prefer to focus on fitting TVP-VAR models using small datasets, with fewer than 20 variables. For instance, in the seminal work by [Primiceri \(2005\)](#), the analysis was limited to a dataset of three variables. More recent papers attempt to overcome these concerns, exploring the use of TVP-VARs on larger datasets ([Koop, 2013](#); [Hauzenberger et al., 2024](#); [Bańbura et al., 2010](#)), through the use of shrinkage and sparsity techniques to restrict the model’s complexity and improve generalizability.

Furthermore, equation 1.3 still assumes homoscedasticity in the error terms, meaning the covariance matrix Σ is static over time. While this assumption simplifies the model, it limits the model’s ability to capture more complex phenomena in the noise terms. To address this, we will introduce stochastic volatility, and allow Σ to evolve over time, capturing changing volatility in key macroeconomic variables like inflation, GDP, and interest rates.

1.2.2 Time-Varying Parameter Regressions

Allowing for time-varying parameters in a VAR model is crucial for capturing the dynamic state of the economy, as it enables contemporaneous changes in the importance of variables. Similarly, this relaxed assumption has also been well studied for univariate regressions ([Kastner and Frühwirth-Schnatter, 2014](#); [Kalli and Griffin, 2014](#); [Belmonte et al., 2014](#); [Bitto and Frühwirth-Schnatter, 2019](#); [Huber et al., 2021](#); [Cadonna et al., 2020](#); [Knaus and Frühwirth-Schnatter, 2023](#)). As we will discuss in section 1.3.1, time-varying parameter regressions (TVP regressions) are a useful framework for estimating TVP-VAR models. We define the TVP regression as a time-varying extension of the univariate regression on y_t in the following way ([Carriero et al., 2019](#); [Cadonna et al., 2020](#)):

$$\begin{aligned} y_t &= X_t \beta_t^i + \varepsilon_t, & \varepsilon_t &\sim N(0, \sigma^2) \\ \beta_t &= \beta_{t-1} + \eta_{i,t}, & \eta_{i,t} &\sim \mathcal{N}(0, \Sigma_B) \end{aligned} \tag{1.4}$$

Where $X_t = (x_{1,t}, \dots, x_{q,t})$ is a row vector of regressors, which may include an intercept term (eg: $x_{1,t} = 1$) or y_t ’s own lags. We define β_t to be the q -dimensional column vector of regressor coefficients at time t , and is the only time-varying parameter for this model. We assume that these coefficients follow a normal random walk with variance-covariance of innovations $\Sigma_B = \text{Diag}(\theta_1, \dots, \theta_q)$, a $q \times q$ diagonal matrix. Finally, we assume a normally distributed error term ε_t .

An alternative representation of the TVP regression decomposes the coefficients β_t into their time-varying and time-invariant portions. This equivalent representation allows for greater control of the time-varying nature of the model. This approach was introduced in [Frühwirth-Schnatter and Wagner \(2010\)](#), and has been used regularly throughout the literature, including many of the papers mentioned above. Following [Frühwirth-Schnatter and Wagner \(2010\)](#), we define the non-centered parameterization of the TVP regression as:

$$\begin{aligned}
y_t &= X_t \beta + X_t \text{Diag} \left(\sqrt{\theta_1}, \dots, \sqrt{\theta_q} \right) \tilde{\beta}_t + \varepsilon_t, & \varepsilon_t &\sim N(0, \sigma^2) \\
\tilde{\beta}_t &= \tilde{\beta}_{t-1} + \tilde{w}_t, & \tilde{w}_t &\sim N(0, I_q) \\
\beta_t &= \beta + \tilde{\beta}_t
\end{aligned} \tag{1.5}$$

Where y_t is a univariate predictor, x_t is a q -dimensional row vector of regressors, β denotes the q -dimensional column vector of time-invariant coefficients, and $\tilde{\beta}_t$ is the corresponding q -dimensional column vector of time-varying coefficients, following a random walk. The innovations of the random walk, $\tilde{w}_t$ are normally distributed with a unit diagonal covariance matrix, implying independent time-variation across coefficients.

1.3 Introducing Stochastic Volatility

1.3.1 Cholesky Decomposition

We build off of equation 1.3 and introduce stochastic volatility by defining a process for the time-variation of the covariance matrix of the errors terms ε_t . We represent this change with $\varepsilon_t \sim \mathcal{N}(0, \Sigma_t)$, introducing a time index t on the covariance matrix of the residuals. We can decompose this matrix in the following way, using the Cholesky Decomposition (Nakajima et al., 2011; Primiceri, 2005; Lubik and Matthes, 2015; Carriero et al., 2019):

$$\Sigma_{u,t} = A_t^{-1} D_t (A_t^{-1})^\top, \tag{1.6}$$

Where $D_t = \text{diag}(e^{d_{1,t}}, \dots, e^{d_{k,t}})$ is a diagonal matrix that contains the time-varying volatility of each variable in y_t , and the structural matrix A_t^{-1} is a $k \times k$ lower triangular matrix with ones on its diagonal. The remaining free (off-diagonal) elements of A_t^{-1} , denoted by $\alpha_{ij,t}, \forall i > j$, are covariance parameters that control the time-varying relationships amongst the error terms of our model (Carriero et al., 2019). The diagonal terms of D_t are assumed to be independent. We formulate a stochastic volatility process on elements of D_t as the following AR(1) process:

$$d_{i,t} = \mu_i + \rho_i(d_{i,t-1} - \mu_i) + w_{i,t}, \quad \text{with } w_{i,t} \sim N(0, \zeta_i^2), \quad \text{for } i = 1, \dots, k \tag{1.7}$$

For each variable i , μ_i is the mean of log variances, $\rho_i \in (-1, 1)$ is an auto-regressive persistence parameter and $w_{i,t}$ are normally distributed error terms with mean 0 and variance ζ_i^2 . We require ρ_i to be bounded between -1 and 1 to ensure stationarity.

The lower triangular structure imposed on A_t^{-1} introduces sensitivity to the ordering of our variables (Chan et al., 2024), and it is possible that any resulting inference or interpretation may change under a different variable ordering. Typically, authors impose restrictions on the structure of this matrix (Primiceri, 2005), citing empirical economic theory to justify their choices. Enforcing strict restrictions on our model seems unnecessary if we can instead allow the data to determine the nature of these restrictions under a Bayesian methodology, and many papers favor this approach (George et al., 2008; Huber and Feldkircher, 2019; Korobilis and Pettenuzzo, 2019).

Despite the issues with the lower-triangular structural matrix of the Cholesky decomposition, this property allows us to estimate a TVP-VAR model using a series of k univariate TVP regressions. To do so, we must incorporate the residual terms of all previous equations as additional regressors of the current equation. This reformulation of the TVP-VAR model is only possible as a result of A_t^{-1} containing no elements above the diagonal. Exploiting this

structure, we can re-write the TVP-VAR model as follows (Carriero et al., 2019; Cadonna et al., 2020):

$$\begin{aligned}
y_{1,t} &= X_t \beta_t^1 + \varepsilon_{1,t}, & \varepsilon_{1,t} &\sim N(0, e^{d_{1,t}}) \\
y_{2,t} &= X_t \beta_t^2 + \alpha_{21,t} \varepsilon_{1,t} + \varepsilon_{2,t}, & \varepsilon_{2,t} &\sim N(0, e^{d_{2,t}}) \\
y_{3,t} &= X_t \beta_t^3 + \alpha_{31,t} \varepsilon_{1,t} + \alpha_{32,t} \varepsilon_{2,t} + \varepsilon_{3,t}, & \varepsilon_{3,t} &\sim N(0, e^{d_{3,t}}) \\
&\dots & & \\
y_{k,t} &= X_t \beta_t^k + \alpha_{k1,t} \varepsilon_{1,t} + \dots + \alpha_{k,k-1,t} \varepsilon_{k-1,t} + \varepsilon_{k,t}, & \varepsilon_{k,t} &\sim N(0, e^{d_{k,t}})
\end{aligned} \tag{1.8}$$

We use the same notation as equation 1.4, with the previously defined AR(1) processes for each log variance $d_{i,t}$. Each β_t^i is a q -dimensional column vector with $q = pk$ and evolves as an independent random walk. In practice, the residuals of the previous $i - 1$ equations $\{\varepsilon_{j,t}\}_{j=0}^{i-1}$ are included as additional regressors in X_t , and the coefficients in β_t^i associated with these residuals correspond with the values of $\alpha_{ij,t}$. We can then reconstruct the matrix A_t^{-1} by appropriately ordering these coefficients, remembering that the diagonal elements are one. This re-formulation of the TVP-VAR model should make it clearer as to why the lower-triangularity of A_t^{-1} causes the order-sensitivity of variables. The literature typically combines the non-centered parameterization of the TVP regression in conjunction with this equation-by-equation approach for estimating TVP-VAR models (Huber et al., 2021; Cadonna et al., 2020).

Reformulating the TVP-SV-VAR model as a series of k TVP regressions was first used by Carriero et al. (2019) in order to avoid repeatedly inverting large matrices, significantly reducing the computational burden of estimation. There are other advantages to this alternative representation, allowing us to estimate TVP-SV-VAR models using existing well-optimized TVP regression packages. We discuss some of these packages in section 3.1, where we address model estimation using R.

1.3.2 Factor Stochastic Volatility

Some papers impose a dimension-reducing factor structure on the data itself (y_t), and proceed with inference by building a VAR or TVP-VAR model on these factors (Nakajima and West, 2013; Lopes and Carvalho, 2007). In a multi-collinear setting, a more effective approach may be to impose a factor structure on the time-varying variance-covariance matrix, as in Kastner and Huber (2020). The factor stochastic volatility model requires the assumption that the stochastic volatility processes for the residuals of y_t are generated by a smaller number $m < k$ of latent volatility factors. We represent the time-varying error covariance matrix Σ_t under a factor structure as follows:

$$\Sigma_t = \Lambda V_t(\Lambda)^\top + \Omega_t \tag{1.9}$$

Where Σ_t is the time-varying $k \times k$ error covariance matrix, Λ is a $k \times m$ matrix of factor loadings, $V_t = \text{Diag}(e^{v_{1,t}}, \dots, e^{v_{m,t}})$ is a $m \times m$ diagonal matrix of time-varying factors and $\Omega_t = \text{Diag}(\sigma_{1,t}^2, \dots, \sigma_{k,t}^2)$ is a $k \times k$ diagonal matrix of idiosyncratic variances not captured by factors in V_t . We update the diagonal elements of V_t and Ω_t with AR(1) processes as follows:

$$\begin{aligned}
v_{i,t} &= \rho_{vi} v_{i,t-1} + w_{vi,t}, & w_{vi,t} &\sim N(0, \zeta_{vi}^2), & i &= 1, \dots, m \\
\log(\sigma_{j,t}^2) &= \mu_{\sigma j} + \rho_{\sigma i} (\log(\sigma_{j,t-1}^2) - \mu_{\sigma j}) + w_{\sigma j,t}, & w_{\sigma j,t} &\sim N(0, \zeta_{\sigma j}^2), & j &= 1, \dots, k
\end{aligned} \tag{1.10}$$

We haven't defined a mean for the volatility process in $v_{i,t}$ since we only wish to capture idiosyncratic noise unrelated to the factors. This type of specification should be reasonable in under multi-collinearity, where we expect at least some of the variables to be highly correlated to each other. Variables like men's unemployment rate and women's unemployment rate are

bound to be highly correlated. We might expect that the variability in both of these time series steams from a single, common, volatility process, justifying a factor approach.

Although we have presented stochastic volatility as an extension of TVP-VAR models, it is relatively straightforward to apply these stochastic volatility specifications to the constant coefficient models. In fact, some research suggests ([Chan and Eisenstat, 2018](#)) that the introduction of time-varying parameters is not nearly as impactful as allowing for heteroscedasticity when it comes to forecasting performance. As stochastic volatility allows for greater flexibility in capturing noise, this versatility may contribute much of the forecasting improvements.

Chapter 2

Estimation and Regularization

2.1 Bayesian vs. Frequentist Approach

VAR, TVP-VAR and TVP-SV-VAR literature typically relies on Bayesian approaches for parameter estimation, as frequentist inference is limited in high-dimensional, non-linear settings. Stochastic volatility, and its associated AR(1) processes, introduce non-linearities and dependence on previous observations, complicating maximum likelihood estimation. While Bayesian methods present their own challenges, they can help deal with the risk of over-parameterization and are capable of handling smaller sample sizes. In contrast, frequentist inference using maximum likelihood estimation requires large sample sizes to establish asymptotic results, making this approach unreliable when the dimension of the parameter space is so much larger than that of the data. Likelihood functions in this setting tend to have multiple peaks, many of which are implausible, further supporting the need for Bayesian methods ([Primiceri, 2005](#)).

Due to high dimensionality, risk of over-parameterization and possible multi-collinearities, regularization techniques are required to ensure accurate forecasting performance. The frequentist approach requires explicit maximization of a penalized likelihood, which could be computationally unfeasible in our setting. Although possible to implement regularization under this regime, the Bayesian approach provides a natural hierarchical framework for introducing regularization. Bayesian priors can induce equivalent shrinkage and sparsity approaches, like the ridge regression and lasso ([Griffin and Brown, 2013](#); [Tibshirani, 1996](#); [Park and Casella, 2008](#)), as well as more flexible forms of regularization discussed in section [2.3](#).

Although the bayesian framework requires us to rigorously define our method for generating point estimates, something which comes naturally for frequentist methods, we gain the ability to perform density forecasts. These density forecasts provide a natural way of quantifying the uncertainty inherent in our model's out-of-sample performance, providing a better understanding of the forecasting distribution. As a result, we will be using Bayesian approaches for all of our needs, including model estimation, regularization and inference. We discuss some of the Bayesian regularization priors and methods in the following section.

2.2 Model Evaluation and Interpretation

2.2.1 Evaluation of Point Forecasts

Although we will work under a Bayesian framework, we can define evaluation criteria for out-of-sample point estimates. We define the root mean square forecast error (RMFSE) as follows (Carriero et al., 2015):

$$RMSFE_h = \sqrt{\frac{1}{k} \sum_{i=1}^k (\hat{y}_{i,t+h} - y_{i,t+h})^2} \quad (2.1)$$

With $\hat{y}_{i,t+h}$ the h -step ahead point forecast for the i -th variable. Although this is typically computed for a single horizon h , we extend this idea and compute the joint root mean square forecast errors as follows:

$$\text{Joint RMSFE} = \sqrt{\frac{1}{H \times k} \sum_{i=1}^k \sum_{h=1}^H (\hat{y}_{i,t+h} - y_{i,t+h})^2} \quad (2.2)$$

This allows us to wholistically evaluate a model's forecasting performance. For both the RMSFE and Joint RMSFE, a lower score is better, as it means that point forecasts are closer to their observed values. We emphasize the distinction between the RMSFE and the Joint RMSFE by specifying whether we are talking about point forecasts *at* horizon h , or *up to* horizon h , respectively.

2.2.2 Predictive Density Scores

Under a Bayesian framework, we can additionally evaluate a model's forecasting performance by determining the accuracy of density forecasts, gaining some understanding of the uncertainty in predictive power. These density forecasts correspond to the posterior predictive densities. We can score the performance of a TVP-VAR model by evaluating the posterior predictive density $p(y_{i,t+h} | y_{1:T})$ at the actual observed value for variable i at horizon h . A model with superior forecasting accuracy should assign more posterior predictive mass to values in the neighborhood of the observed value, leading to a higher density score. This approach provides a framework for model comparison that takes into account the shape of the forecasting distribution.

Generating samples from the posterior predictive distribution, especially at larger time horizons, is generally not trivial, especially in our high-dimensional setting. To illustrate, we present the integral from computing the 1-step ahead posterior predictive distribution:

$$p(y_{t+1} | y_{1:T}) = \int p(y_{i,t+1} | \theta, y_{1:T}) p(\theta | y_{1:T}) d\theta \quad (2.3)$$

Where θ represents all model parameters, and $y_{1:T}$ denotes all observations from time $t = 1, \dots, T$. Integrating out the posterior distribution of our parameters is exceedingly difficult to the high-dimensionality of the parameter space for TVP-VAR models. At larger forecasting horizons this issue is compounded, as it is necessary to marginalize the intermediate forecasting horizons between $y_{i,t+1}$ and $y_{i,t+h}$ before arriving to our desired h -step ahead posterior predictive distribution $p(y_{i,t+h} | y_{1:T})$. Even with an MCMC approach, this computation may be infeasible

due to the size of our model. Instead, [Carriero et al. \(2015\)](#) proposes a quadratic approximation to the log predictive density for variable i at horizon h , inspired by [Adolfson et al. \(2007\)](#):

$$s(y_{i,t+h}) = -\frac{1}{2} \left(\ln(2\pi) + \ln(V_{i,t+h|t}) + \frac{(y_{i,t+h} - \hat{y}_{i,t+h|t})^2}{2V_{i,t+h|t}} \right) \quad (2.4)$$

Where $\hat{y}_{i,t+h|t}$ and $V_{i,t+h|t}$ are the posterior mean and variance of the simulated forecast distribution for variable i at h steps ahead, respectively. Instead of integrating out all intermediate forecasts, we generate a large number of point forecasts, compute their variance and mean, and use these to approximate the log predictive density. Then, we compute the above approximation of the log predictive density using the observed value of $y_{i,t+h}$.

As presented by [Carriero et al. \(2015\)](#), these log predictive scores are computed for a single variable at a single horizon. We adapt this evaluation criteria and average the log predictive scores over all variables at the same forecasting horizon, as follows:

$$\text{LPDS}_h = \frac{1}{k} \sum_{i=1}^k s(y_{i,t+h}) \quad (2.5)$$

We denote this the average log predictive density score (LPDS) at horizon h . Although just an approximation, a higher LPDS should be indicative of a more appropriate posterior predictive distribution for our out-of-sample data, without necessitating the computation of intractable integrals.

2.2.3 Impulse Response Functions

By using VAR and TVP-VAR models on economic data, we hope to gain insight on the relationship a shock in one variable, say GDP, might have on some or all of the remaining variables, such as inflation or unemployment. To understand how the noise in our variables impact others, one might consider inspecting the posterior point estimates of the coefficient matrices, since these should already encode how lag p of variable j has an impact on $y_{i,t}$. A more mathematically sound approach uses these matrices to compute the forecast error impulse responses (FEIR), as these more directly allow us to understand how forecasting errors propagates throughout the remaining variables. The FEIR defined at horizon h is defined as follows (as in [Primiceri \(2005\)](#); [Lütkepohl \(2005\)](#)):

$$\Phi_{h,t} = \sum_{i=1}^p \Phi_{h-i,t} B_{i,t}, \quad \text{for } j = 1, \dots, h \quad (2.6)$$

With $\Phi_{0,t} = I_k$ the $k \times k$ identity matrix. Due to the construction of the TVP-VAR model, the residuals ε_t are likely to be correlated across variables, which complicates the interpretation of the error propagation. Contemporaneous correlations in the noise term (the covariance terms of Σ_t) impede our understanding of how shocks propagate throughout our model. These correlations justify the introduction of the structural matrix A_t^{-1} , which helps us to disentangle these correlations, interpret them, and impose independent AR(1) processes for the elements of D_t . We can ensure independent processes in D_t by noting that the residuals ε_t can be equivalently re-written as $\varepsilon_t = A_t^{-1} D_t^{1/2} \xi_t$ with $\xi_t \sim \mathcal{N}(0, I_k)$. As ξ_t has no time variation and has i.i.d elements, we can guarantee that elements of D_t are independent, which is only possible

because A_t^{-1} is capturing correlation of the errors. We derive structural impulse responses from the FEIRs $\Phi_{i,t}$, which should be devoid of these correlations, as follows:

$$\Theta_{j,t} = \Phi_{j,t} A_t^{-1} \quad (2.7)$$

For a constant coefficient VAR model (especially without stochastic volatility), estimation of both impulse responses is straightforward, only a matter of recursively computing the product of our matrices. For TVP-VAR and stochastic volatility models, computing impulse responses is generally more involved, as we need to sample a new coefficients at each horizon, and use these to calculate impulse responses. In general, MCMC methods are required to compute impulse responses.

2.3 Choice of Prior and Regularization Priors

Although it is possible to construct a conjugate prior for the TVP-SV-VAR model, this approach fails to yield efficient results when dealing with larger datasets. Because conjugacy does not introduce regularization, these types of priors are too restrictive for our setting, as they cannot handle multi-collinearities. We therefore require a richer family of distributions and priors to obtain improved inference (Carriero et al., 2015).

We use the following sections to discuss a variety of regularization priors that are well explored in the VAR and TVP-VAR literature, how they differ in their approaches to shrinkage and sparsity, and why their properties are useful given our dataset. The aim of these regularization priors is to decrease the sensitivity of our model to noise, avoid over-fitting, and hopefully improve performance and interpretability of the results.

For simplicity, we introduce these regularisation priors in the context of TVP regressions, as we build upon them for fitting TVP-VAR models. As we introduce these models, we will be using notation referring to the time-invariant coefficients of the non-centered TVP regression, β_i . These priors can be easily extended to the time-varying case by applying them to the standard deviation of the random walk innovation $\text{Diag}(\sqrt{\theta_1}, \dots, \sqrt{\theta_q})$, making sure to use separate parameters and hyperparameters for each group of coefficients.

We include additional information regarding the full conditional posterior of these priors in Appendix A.

2.3.1 Bayesian Lasso

We have previously defined the lasso as a constrained optimization problem. However, this formulation is better suited for frequentist methods, requiring the computation of a point estimate $\hat{\beta}_{lasso}$ that minimizes the loss function. We introduce the equivalent bayesian priori for this optimization problem. The lasso estimator for a regression coefficient vector, β , can be considered to be the maximum a-posteriori draws of independently distributed Laplace priors (Tibshirani, 1996; Park and Casella, 2008). The Laplace (double exponential) distribution for a scalar random variable, has the following density, with a location parameter μ , and a scale parameter b :

$$f(x \mid \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right) \quad (2.8)$$

As shown in [Park and Casella \(2008\)](#) we can represent this as a normal-exponential mixture, which allows us to impose a hierarchical structure on the coefficient priors. Using this normal-exponential mixture, [Belmonte et al. \(2014\)](#) builds on this prior to construct the bayesian lasso prior for time-varying parameter models. We define the bayesian lasso for coefficient β_i as ([Park and Casella, 2008](#); [Belmonte et al., 2014](#); [Korobilis and Shimizu, 2022](#)):

$$\beta_i | \tau_i^2 \sim N(0, \tau_i^2), \quad \tau_i^2 | \lambda \sim \text{Exp}\left(\frac{\lambda^2}{2}\right) \quad (2.9)$$

Here, λ is a parameter that controls the overall (global) shrinkage for each set of parameters. For a full bayesian approach, one may assign a hyperprior on λ^2 . For example, in [Belmonte et al. \(2014\)](#), a $\text{Gamma}(0.001, 0.001)$ prior is used to create a fully hierarchical bayesian lasso prior. Choosing small values for the hyperprior's parameters implies an uninformative prior on the shrinkage parameter, allowing the data to determine how much regularization is necessary.

The bayesian lasso has proven to be quite an effective tool for variable selection ([Park and Casella, 2008](#); [Belmonte et al., 2014](#)), since the Laplace distribution allows for an extreme and sharp peak at zero (indicating sparsity), as well as boasting heavy tails when the coefficients are non-zero. Under multi-collinearity, variable selection is necessary to avoid overfitting as the inclusion of an additional highly correlated variable is unlikely to increase predictive. As [Bitto and Frühwirth-Schnatter \(2019\)](#) shows, the bayesian lasso makes great use of the non-centered TVP regression, by allowing the data to determine whether a coefficient should be shrunk to exactly zero ($\beta_i = 0$), time-varying ($\sqrt{\theta_i} \neq 0$), or time-invariant ($\sqrt{\theta_i} = 0$). These parameter outcomes make the bayesian lasso a powerful variable selection prior for TVP models.

2.3.2 Stochastic Search Variable Selection

Stochastic Search Variable Selection (SSVS) is a bayesian methodology for implementing variable selection through the use of spike and slab priors. These spike and slab priors introduce sparsity through a mixture prior that assigns coefficients to be shrunk towards zero, or remain entirely unaltered. Spike and slab priors as they first introduced by [Mitchell and Beauchamp \(1988\)](#) use dirac-delta functions to impose exact sparsity. A more modern approach by [George et al. \(2008\)](#) allows the prior to remain continuous. For a constant coefficient VAR(1) model, we define the continuous spike and slab prior for coefficient B_i as:

$$B_i | \gamma_i \sim (1 - \gamma_i)N(0, \sigma_i^2 \tau_{0,i}^2) + \gamma_i N(0, \sigma_i^2 \tau_{1,i}^2), \quad \text{for } i = 1, \dots, k^2 \\ \gamma_i \sim \text{Bernoulli}(p) \quad (2.10)$$

Where $\tau_{0,i}^2 \ll \tau_{1,i}^2$. The idea is that we can use draws from γ_i to determine the variance of coefficient B_i . If $\gamma_i = 1$, indicating a non-sparsity, we allow the coefficient to be unrestricted. Otherwise, if $\gamma = 0$, we force the parameter's variance to be significantly smaller, effectively sparsifying by making the coefficient negligible. Variable selection is determined entirely by γ_i . Although the continuous formulation does not induce exact sparsity, with appropriate choices for $\tau_{0,i}$ and $\tau_{1,i}$ we can achieve a similar result. In practice, we may also want to introduce a beta distribution hyperprior on the inclusion probability p , and an appropriate distribution for the variances σ_i^2 .

SSVS allows us to succinctly model uncertainty in the predictive power of different variables in our model. It allows us to fit unrestricted coefficients while still imposing sparsity. With a hyperprior on p , we can allow the data to determine appropriate inclusion probabilities. SSVS should be quite effective for forecasting and impulse response analysis, as the MCMC draws of forecasted values for $y_{i,t+h}$ will be averaged over variable-inclusion uncertainty.

2.3.3 Global-Local Shrinkage

Many of the most interesting priors for our setting are the so-called ‘global-local shrinkage’ priors (Polson and Scott, 2011), which we will explore in the following sections. These priors share the characteristic of introducing a global shrinkage parameter, τ , that controls the overall level of shrinkage, along with local parameters, ϕ_i , for $i = 1, \dots, q$, which determine the degree of shrinkage for each variable, individually. The purpose of these two groups of parameters is to provide more granular control over the regularization applied to the model, allowing us to shrink non-relevant predictors toward zero while retaining flexibility for important ones. For example, the Bayesian lasso is considered a global shrinkage estimator, as it applies shrinkage uniformly. Although global-local shrinkage priors introduce more parameters, they offer a valuable enhancement by providing finer control over our regularization approach.

2.3.4 Horseshoe Prior

The horseshoe estimator was originally proposed as a prior for sparse signals in Carvalho et al. (2010), helping to separate between noisy variables and those with more significant predictive power. We first define the half-cauchy distribution (with location parameter μ and scale parameter c) and immediately introduce the horseshoe prior density for coefficient β_i :

$$f_{C^+}(x; \mu, c) = \frac{2}{c\pi} \frac{1}{1 + [(x - \mu)^2/c^2]} \text{ for } x > \mu \quad (2.11)$$

$$\beta_i | \lambda_i \sim \mathcal{N}(0, \sigma^2 \lambda_i^2 \tau^2), \quad \lambda_i \sim C^+(0, 1), \quad \tau \sim C^+(0, 1) \quad (2.12)$$

Where $C^+(\mu, c)$ denotes the density in equation 2.11. Although we have specified a half-cauchy prior for τ , this is more so a choice typical in the literature Follett and Yu (2019); Bhadra et al. (2020); Korobilis and Shimizu (2022) rather than a stringent necessity, with any appropriate prior ($\tau \in \mathbb{R}^+$) retaining the same properties. The horseshoe induces global-local shrinkage, with λ_i controlling coefficient-specific shrinkage, and τ controlling the overall model shrinkage. The horseshoe prior is also a continuous shrinkage prior, with a sharp peak around zero and heavy tails. Although this means that the horseshoe prior will not set any of the coefficients exactly to zero, coefficients that are not strongly supported by the data will be more aggressively shrunk towards zero. Due to the half-cauchy priors, the horseshoe has heavy, allowing for relatively unrestricted coefficients if the data supports non-sparsity.

2.3.5 Normal-Gamma Prior

Like the horseshoe, the normal-gamma prior is a continuous global-local shrinkage prior. The normal-gamma’s ability to induce shrinkage and perform variable selection using a straightforward hierarchy has made it a well-studied prior in the literature. We introduce the normal-gamma prior as in Brown and Griffin (2010):

$$\beta_i \sim \mathcal{N}(0, \phi_i), \quad \phi_i \sim \text{Gamma}(\lambda, \gamma) \quad (2.13)$$

Here, λ and γ control global shrinkage whereas ϕ_i controls local shrinkage. Like the horseshoe prior, the normal-gamma does not induce exact sparsity, but can heavily shrink non-significant coefficients close to zero, making it a powerful tool for both variable selection and shrinkage. The normal-gamma offers significantly more flexibility in prior shapes due to the

gamma distribution's relationship with other distributions, allowing for a greater breadth of prior specifications, even eliciting some of the previously discussed priors as special cases. For example, by setting $\lambda = \frac{1}{2}$ and $\gamma = \frac{\alpha}{2}$, we recover the bayesian lasso with shrinkage parameter α . Additionally, the normal-gamma prior belongs to the family of scale mixture of normals, of which the spike and slab prior is also a member. With a careful choice for λ , it is possible to use the normal-gamma prior to recover the spike and slab priors in equation 2.10 (Brown and Griffin, 2010). Other scale mixture of normals can also be represented by introducing additional priors to the normal-gamma hierarchy, increasing heavy-tailedness (Griffin and Brown, 2017; Bitto and Frühwirth-Schnatter, 2019; Huber et al., 2021). We mention these additional hierarchies in the following section.

2.3.6 Triple-Gamma Prior

The triple-gamma prior is a continuous global-local shrinkage prior, meant to induce regularization on the variance (θ_i) of random walk innovations for time-varying coefficients in a non-centered TVP regression. Due to the flexibility of the triple-gamma, it also lends itself to an equivalent representation using a normal-gamma-gamma prior, a hierarchical extension of the normal-gamma. We follow Cadonna et al. (2020) to define the triple-gamma prior on θ_i as follows:

$$\theta_i | \xi_i^2 \sim \mathcal{G}\left(\frac{1}{2}, \frac{1}{2\xi_i^2}\right), \quad \xi_i^2 | a^\xi, \kappa_i^2 \sim \mathcal{G}\left(a^\xi, \frac{a^\xi \kappa_i^2}{2}\right), \quad \kappa_i^2 | c^\xi, \kappa_B^2 \sim \mathcal{G}\left(c^\xi, \frac{c^\xi}{\kappa_B^2}\right) \quad (2.14)$$

Where a^ξ , c^ξ and κ_B^2 are hyperparameters to be carefully fixed. These hyperparameters control how much mass we want to assign to values near zero, how heavy we want the tails to be and how much global shrinkage we want to induce, respectively (Cadonna et al., 2020; Knaus and Frühwirth-Schnatter, 2023). Smaller values of c^ξ lead to heavier tails, while larger values for κ_B^2 causes more shrinkage. The triple-gamma prior on θ_i induces shrinkage on the amount of time variation in the coefficients. Shrinkage towards zero, in this context, implies fixed, time-invariant coefficients, inducing variance selection. Cadonna et al. (2020) shows that a triple-gamma prior on θ_i is equivalent to a normal-gamma-gamma on $\sqrt{\theta_i}$ and uses a normal-gamma hierarchical extension to introduce the following prior on both $\sqrt{\theta_i}$ and the time-invariant coefficients:

$$\begin{aligned} \sqrt{\theta_i} | \xi_i^2 &\sim N(0, \xi_i^2), \quad \xi_i^2 | a^\xi, \kappa_i^2 \sim \mathcal{G}\left(a^\xi, \frac{a^\xi \kappa_i^2}{2}\right), \quad \kappa_i^2 | c^\xi, \kappa_B^2 \sim \mathcal{G}\left(c^\xi, \frac{c^\xi}{\kappa_B^2}\right) \\ \beta_i | \tau_i^2 &\sim \mathcal{N}(0, \tau_i^2), \quad \tau_i^2 | a^\tau, \lambda_i^2 \sim \mathcal{G}\left(a^\tau, \frac{a^\tau \lambda_i^2}{2}\right), \quad \lambda_i^2 | c^\tau, \lambda_B^2 \sim \mathcal{G}\left(c^\tau, \frac{c^\tau}{\lambda_B^2}\right) \end{aligned} \quad (2.15)$$

Where a^τ , c^τ , λ_B^2 , a^ξ , c^ξ and κ_B^2 are hyperparameters, as above. This prior forms an extremely rich family of prior shapes and specifications, with a significant number of important priors arising as special cases, including the lasso, horseshoe and normal-gamma priors. Table 1 in Cadonna et al. (2020) shows many of the different, well studied priors that arise as special cases of the triple-gamma. We include a table with these special cases in Appendix B. This flexibility comes with many benefits, including the ability to perform global-local shrinkage, time-variation uncertainty average, variable selection and variance selection.

Chapter 3

Data and Methodology

3.1 Exploring the Data

Many variable choices in our data set are inspired by [Carriero et al. \(2015\)](#). Their paper discusses transformations of the data prior to model estimation, and other considerations. Our chosen data should provide a relatively comprehensive outlook of the macro-economy, while still being concise enough to enable model estimation in a reasonable timeframe. The number of parameters of a TVP-SV-VAR model grows quadratically in the size of the dataset and thus the quantity of variables must be chosen carefully. Other variables, such as a housing index, savings rate, and employment wages were also considered, but ultimately deemed unnecessary.

Table 3.1 provides a list of the included variables, how the data was seasonally adjusted, the transformations performed, and the corresponding St. Louis Federal Reserve Database (FRED) series code. We choose to work with macro-economic data regarding only the United States due to ease of access to a large variety of time series.

FRED Code	Series	Seasonally Adjust	Transformation
WFRBST01134	Share of Total Net Worth Held by the Top 1% (99th to 100th Wealth Percentiles)	FRED	None
WFRBSB50215	Share of Total Net Worth Held by the Bottom 50% (1st to 50th Wealth Percentiles)	FRED	None
MCOILWTICO	Crude Oil Prices: West Texas Intermediate (WTI) - Cushing, Oklahoma	Manual	$400 * \ln(y_t / y_{t-1})$
T10Y3MM	10-Year Treasury Constant Maturity Minus 3-Month Treasury Constant Maturity	Manual	$400 * \ln(y_t / y_{t-1})$
MICH	University of Michigan: Inflation Expectation	Manual	$400 * \ln(y_t / y_{t-1})$
T10Y2YM	10-Year Treasury Constant Maturity Minus 2-Year Treasury Constant Maturity	FRED	None
M2SL	M2 Money Supply	FRED	$400 * \ln(y_t / y_{t-1})$
PCECC96	Real Personal Consumption Expenditures	FRED	$400 * \ln(y_t / y_{t-1})$
AAAFFM	Corporate Bond Yield Minus Federal Funds Rate	Manual	None
AAA10YM	Corporate Bond Yield Relative to Yield on 10-Year Treasury Constant Maturity	Manual	None
UMCSENT	University of Michigan: Consumer Sentiment	Manual	None
GDPIC1	Real Gross Domestic Product	FRED	$400 * \ln(y_t / y_{t-1})$
INDPRO	Industrial Production: Total Index	FRED	$400 * \ln(y_t / y_{t-1})$
PPIACO	Producer Price Index by Commodity: All Commodities	Manual	$400 * \ln(y_t / y_{t-1})$
DPCERD3Q086SBEA	Personal consumption expenditures (implicit price deflator)	FRED	$400 * \ln(y_t / y_{t-1})$
A822RD3Q086SBEA	Government consumption expenditures and gross investment (implicit price deflator)	FRED	$400 * \ln(y_t / y_{t-1})$
OPHNFB	Nonfarm Business Sector: Labor Productivity (Output per Hour) for All Workers	FRED	$400 * \ln(y_t / y_{t-1})$
NASDAQCOM	NASDAQ Composite Index	Manual	$400 * \ln(y_t / y_{t-1})$
FEDFUNDS	Federal Funds Effective Rate	FRED	None
LNS14000001	Unemployment Rate - Men	FRED	None
LNS14000002	Unemployment Rate - Women	FRED	None
LNS14000003	Unemployment Rate - White	FRED	None
LNS14000006	Unemployment Rate - Black or African American	FRED	None
LNS14000009	Unemployment Rate - Hispanic or Latino	FRED	None

Table 3.1: Description of dataset and transformations. FRED seasonal adjustment indicates that data was downloaded with seasonal adjustment from the St. Louis Federal Reserve database, while Manual indicates seasonal adjustment with the `seasonal` R package.

Some of the data was gathered from FRED already seasonally adjusted. If the seasonal

adjustment wasn't publicly available, we used the `seasonal` R-package (Sax and Eddelbuettel, 2018) which uses the same methodologies as FRED. Previous work on TVP-SV-VAR models that uses monthly data typically applies the transformation $1200 * \ln(y_t/y_{t-1})$, representing the variables as continuously compounding annual rates of change (Carriero et al., 2015). Instead, we use quarterly data starting 1989Q4 and ending 2024Q1, with the equivalent transformation being $400 * \ln(y_t/y_{t-1})$. For some variables, log transformations are necessary to ensure stationarity. Log differences were taken for as many variables as possible, avoiding those with negative values to prevent NA values. Variables like the Federal Funds rate, the unemployment rate, and wealth distribution data are intentionally untransformed, due to their importance as explanatory economic variables and naturally bounded nature.

Although the shrinkage priors we've introduced are helpful for dealing with multi-collinearities, we take additional steps to ensure they do not hinder computational inference. When fitting our models, we divide each variable in our dataset by their corresponding standard deviations. We do not subtract the means to retain information about the intercept terms. We mimic the literature (Korobilis, 2013; Carriero et al., 2019; Kastner and Huber, 2020) in this choice, as it allows us to better interpret the impulse responses and helps prevent against the possibility of computationally singular matrices.

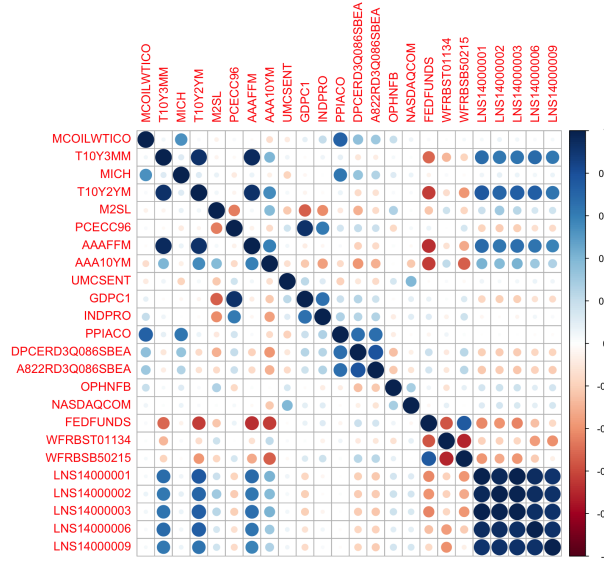


Figure 3.1: Heatmap of the Correlation Matrix for our Main Dataset, displaying the correlations present in our dataset.

We display a heatmap of the correlation matrix of our dataset in figure 3.1 to highlight multi-collinearities. Not all variables are correlated with each other, but subgroups of variables appear to be. For example, all demographic-specific unemployment data is highly correlated with each other, as well as with a variety of treasury and credit spreads (differences between the interest rate of treasuries with different maturities). We use the correlation matrix and principal components analysis (PCA) to understand the collinearities, showing why the standard TVP-VAR without shrinkage priors performs poorly.

To fit a VAR model with factor stochastic volatility, we must determine the number of factors to use. Since many exploratory factor models are based on a principal component approach, we present the proportion of variance explained by the principal components of our data. Table 3.2 displays the proportion of variance explained for the first 12 principal components. Twelve principal components are required to explain $> 95\%$ of the variance in our data, and this becomes our choice for the number of factor stochastic volatility components.

Component no.	1	2	3	4	5	6	7	8	9	10	11	12
Cumulative % of var. explained	0.3187	0.4739	0.5860	0.6906	0.7640	0.8087	0.8512	0.8812	0.9059	0.9271	0.9459	0.9619

Table 3.2: Table Showing Cumulative Proportion of Variance Explained derived from PCA on our un-scaled dataset

3.2 Gibbs Sampling for TVP-SV-VARs

Because many of our priors can be elicited as special cases of the triple-gamma, we discuss the MCMC algorithm required to generate samples from a TVP-SV-VAR by from samples of each TVP regression, as in equation 1.8. Details on how to sample from a non-centered TVP regression with triple-gamma priors can be found in Algorithm 1 of [Cadonna et al. \(2020\)](#), which we will outline in Appendix B for reference. The `updateTVP` function of the `shrinkTVP` package implements this algorithm.

The algorithm proposed by [Cadonna et al. \(2020\)](#) to estimate a TVP-SV-VAR model using TVP regressions is presented as follows:

1. Initialize values for the structural matrix A_t^{-1} , the standard deviation of the random walk innovations for elements of A_t^{-1} and all fixed hyperparameters.
2. For each equation i : Initialize starting values for $\beta^{(i)}$, the global and local shrinkage parameters for $\beta^{(i)}$, $\tilde{\beta}_0^{(i)}$, the standard deviations of the innovations $\{\sqrt{\theta_j}\}_{j=0}^q$, and the global and local shrinkage parameters for $\{\sqrt{\theta_j}\}_{j=0}^q$, the log volatilities $d_{i,0}$ and the corresponding volatility process parameters μ_i , ρ_i , and ζ_i^2 .
3. For each equation $i = 1, \dots, k$: Conditional on A_t^{-1} and $D_t = \text{Diag}(e^{d_{1,t}}, \dots, e^{d_{k,t}})$, generate samples from the conditional posterior of β^i , θ_j and their global and local shrinkage parameters. With $\beta_t^{(i)} = \beta^{(i)} + \tilde{\beta}_t^{(i)}$, we generate these samples for the following regression:

$$y_{i,t} = c_t + X_t \beta_t^{(i)} + \sum_{j=1}^{i-1} \alpha_{ij,t} \varepsilon_{j,t} + \varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim N(0, e^{d_{i,t}}) \quad (3.1)$$

4. For each equation $i > 1$: Conditional on $\beta_t^{(i)}$, generate samples from the following TVP regression for the the elements of A_t^{-1} , the parameters associated with their time variation, and the log volatilities:

$$y_{i,t}^* = y_{i,t} - X_t \beta_t^{(i)}, \quad y_{i,t}^* = \sum_{j=1}^{i-1} \alpha_{ij,t} \varepsilon_{j,t} + \varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim N(0, e^{d_{i,t}}) \quad (3.2)$$

Sampling of stochastic volatility processes in this setting typically uses the algorithm proposed by [Kim et al. \(1998\)](#) for AR(1) processes. This algorithm is implemented in the `stochvol` and `factorstochvol` R packages ([Kastner, 2016](#); [Hosszejni and Kastner, 2021](#)) and thus does not require further detailed discussion.

3.3 Models To Be Fit

R packages like `vars`, `BVAR`, `bvarsv`, `bvartools`, `bayesianVARs` and `shrinkTVPVAR` allow for out-of-the-box estimation of constant coefficient and TVP-VAR models with a variety of

different shrinkage priors and stochastic volatility specifications. The `shrinkTVP` package allows for posterior sampling of TVP regressions with for triple-gamma, normal-gamma and ridge regressions, which can be used with algorithm 3.2 to generate samples of a TVP-VAR model with shrinkage. Using special cases of the triple-gamma we can generate samples for hierarchical horseshoe and lasso priors. In this section, we will discuss the models to be estimate, and which packages were used. We list these models below, and reference these models in the future using the bolded acronyms:

1. **CC-VAR**: Constant Coefficient VAR model without Stochastic Volatility. Estimated using `vars` with OLS-type estimators.
2. **VAR-SV-N**: CC-VAR with stochastic volatility and normal priors on the coefficients.
3. **VAR-SV-NG**: CC-VAR with Cholesky stochastic volatility and a normal-gamma shrinkage prior on the coefficients, using the `bayesianVARs` package.
4. **VAR-SV-HS**: CC-VAR with cholesky stochastic volatility and a horseshoe prior on the coefficients.
5. **VAR-SV-SSVS**: CC-VAR with cholesky stochastic volatility and an SSVS prior on the coefficients.
6. **VAR-FSV**: CC-VAR with 12-Factor Stochastic Volatility.
7. **VAR-FSV-NG**: VAR-FSV (12 factors) with normal-gamma shrinkage priors on the coefficients.
8. **VAR-FSV-HS**: VAR-FSV (12 factors) with horseshoe shrinkage priors on the coefficients.
9. **TVP-SV-VAR-BL**: TVP-SV-VAR with cholesky stochastic volatility and a Hierarchical Bayesian Lasso prior on the model coefficients and coefficient innovation standard deviations.
10. **TVP-SV-VAR-HS**: TVP-SV-VAR with a Hierarchical Horseshoe Prior on the model coefficients and coefficient innovation standard deviations.
11. **TVP-SV-VAR-NG**: TVP-SV-VAR with a Normal-Gamma prior on the model coefficients and coefficient innovation standard deviations.
12. **TVP-SV-VAR-TG**: TVP-SV-VAR with a Triple-Gamma prior on the model coefficients and coefficient innovation standard deviations.

We use the `vars` and `factorstochvol` packages (Hosszejni and Kastner, 2021) for constant coefficient stochastic volatility models. We use the `shrinkTVP` and `shrinkTVPVAR` package (Knaus et al., 2021; Knaus, 2024) for all TVP models. The `shrinkTVPVAR` package is still under development, but the author, Dr. Peter Knaus, was gracious enough to share a working-version of the package, and should be widely available in the near future.

Chapter 4

Results and Discussion

The main motivation of this paper is to discuss the effect of monetary policy and macroeconomic shocks on dis-aggregated unemployment and wealth distribution data. We focus our discussion on results involving the Federal Funds Rate, Unemployment Rates, and Share of Total Net Worth variables. Because we have quarterly data, we fit all models with $p = 4$ lags, representing a calendar year’s worth of information. We do not fit factor stochastic volatility models on the lasso, SSVS, normal-gamma or triple-gamma priors, as they already encode notions of variable or variance selection.

To generate posterior samples, we use the Gibbs Sampler to generate 5,000 posterior draws after 5,000 burn-in draws, for 10,000 total iterations, and no thinning. Not all of the data was used for generating posterior draws, reserving the final 20 observations (5 years) to evaluate forecasting performance. Evaluation of our models uses posterior mean forecasting point estimates to compute the RMSFE and Joint RMSFE up to time $T + h$, for horizons $h = 1, 2, 3, 4, 8, 12, 16, 20$. This is in contrast to much of the literature, which uses smaller forecasting horizons (Belmonte et al., 2014; Koop, 2013; Bitto and Frühwirth-Schnatter, 2019). We also compute average log predictive density scores using these values for h . Log predictive scores are computed for all models except the CC-VAR, as this uses a frequentist estimator. We present these results in Tables 4.1, 4.2 and 4.3.

When computing impulse response functions, we introduce a 1 standard deviation shock to our impulse variable and propagate this shock up to 20 quarters in the future. To examine and interpret the impulse responses, we use the $[0.16, 0.84]$ credible intervals for the MCMC samples of the responses, which is standard in the literature (Primiceri, 2005; Cogley and Sargent, 2005). This credible interval corresponds to the bounds of a 1 standard deviation credible interval under a normal distribution.

4.1 Forecasting Errors

The h -step ahead RMSFE (Table 4.1) for constant-coefficient VARs with shrinkage performs better, in general, than for the CC-VAR without stochastic volatility using an OLS-type estimator, which we expected. However, comparing the RMSFE for the VAR-SV-N with a normal prior against those with shrinkage priors indicates that the increased performance of stochastic volatility is negligible without additional regularization. Similarly, for joint RMSFE results in table 4.2, constant coefficient VARs with both stochastic volatility and shrinkage seem to perform better on average than the vanilla VAR, but stochastic volatility alone is not enough to improve forecasting. Although the VAR-SV-N possesses a greater ability to deal with noise, the

normal prior impedes further improvements under a multi-collinear setting. Factor stochastic volatility specifications, even with normal priors, perform quite well in comparison. This result supports the belief that a smaller number of volatility processes are generating the variance in our data, the main assumption of factor stochastic volatility approaches.

Amongst TVP models, the normal-gamma prior shows improved forecasting over the lasso, triple-gamma or horseshoe, though it still falls short compared to constant coefficient models at longer horizons. The underperformance of most of our TVP-SV-VAR models may not be unexpected. The triple-gamma prior, for example, requires a significant number of parameters. The time-invariant coefficients, the process innovation variances of the time-varying coefficients, the elements of the structural matrix and their process innovation variances — all require their own set of 4 global and 4 local shrinkage parameters. Thus, there are at least 16 global shrinkage parameters and 16 local shrinkage parameters for each variable, of which we have 24. With a relatively small number of observations in the dataset and a high parameterization, we can only assume that 10,000 gibbs iterations are not sufficient to ensure convergence to the correct posterior.

In Cadonna et al. (2020), 100,000 posterior draws were generated, with an additional 100,000 used for burn-in. Due to time and computing constraints, generating such a large number of draws was not possible. A MacBook Pro with a 14-core M3 Max processor and 36 GB of RAM was used for model estimation. Each TVP-SV-VAR required approximately 6 hours of compute for 10,000 total gibbs sampler draws. Extrapolating, generating 200,000 draws could require 5 days of computing time, per model.

Another possibility is that our chosen metrics may not be appropriate for TVP models. For each forecasting horizon, the `shrinkTVP` and `shrinkTVPVAR` packages generates a forecast sample for each non-burn-in draw, at each horizon. In our case, this would correspond with 5,000 draws per horizon, per variable. Computing RMSFE with the posterior mean of these forecasts and then computing the RMSE over all variables may be cumulatively adding errors, causing our metrics to become unusually large. When using the posterior median of the forecasts, the TVP-SV-VAR-BL gains an improvement, but other models do not. We include these values in Appendix D, as well as density plots for underperforming TVP-VAR models, to support the idea of non-convergence in the gibbs sampler. Code for computing the RMSFE is available in Appendix C.

	h -step ahead RMSFE							
h	1	2	3	4	8	12	16	20
CC-VAR	50.24	83.15	35.81	39.35	85.16	30.82	63.17	8.83
VAR-FSV-HS	7.69	7.55	18.25	62.9	27.47	34.69	17.8	17.77
VAR-FSV-NG	9.44	8.4	16.04	64.83	28.98	37.42	16.45	18.09
VAR-SV-HS	8.59	7.16	18.02	63.36	28.51	38.83	18.48	15.11
VAR-SV-N	45.87	56.69	74.53	130.53	304.57	1.71e+03	1.51e+04	1.22e+05
VAR-SV-NG	10.92	9.06	15.84	64.98	27.86	37.24	18.88	13.73
VAR-SV-SSVS	9.34	6.41	17.71	63.52	28.15	38.28	18.28	14.97
TVP-SV-VAR-BL	892.58	5.25e+09	2.27e+16	8.68e+22	9.34e+50	2.79e+80	4.24e+109	1.38e+139
TVP-SV-VAR-HS	7.24e+05	3.60e+08	7.98e+12	2.44e+18	2.00e+39	2.08e+61	3.90e+82	5.06e+105
TVP-SV-VAR-NG	18.67	7.3	11.24	70.95	37.71	145.27	9.06e+03	1.17e+06
TVP-SV-VAR-TG	2.02e+04	2.40e+06	4.46e+06	6.64e+10	1.80e+19	5.60e+28	3.15e+39	2.03e+50

Table 4.1: RMSFE results for h step ahead forecast for each model.

	Joint RMSFE up to horizon h						
h	2	3	4	8	12	16	20
CC-VAR	64.48	53.99	50.13	65.61	59.55	57.8	54.06
VAR-FSV-HS	7.62	12.23	33.19	30.37	28.39	27.5	25.89
VAR-FSV-NG	8.93	11.79	33.98	30.78	30.43	29.61	31.33
VAR-SV-HS	7.91	12.24	33.41	30.52	28.9	27.77	26.03
VAR-SV-N	51.57	60.2	83.53	168.82	811.0	5.43e+03	3.54e+04
VAR-SV-NG	10.03	12.28	34.18	30.76	28.91	27.87	25.97
VAR-SV-SSVS	8.01	12.14	33.45	30.6	28.87	27.8	26.07
TVP-SV-VAR-BL	3.71e+09	1.31e+16	4.34e+22	3.30e+50	8.05e+79	1.06e+109	3.09e+138
TVP-SV-VAR-HS	2.55e+08	4.61e+12	1.22e+18	7.08e+38	6.01e+60	9.74e+81	1.13e+105
TVP-SV-VAR-NG	14.18	13.27	37.29	33.48	53.3	2.38e+03	2.73e+05
TVP-SV-VAR-TG	1.70e+06	2.92e+06	3.32e+10	6.38e+18	1.62e+28	7.88e+38	4.55e+49

Table 4.2: Joint RMSFE results up to forecasting horizon h for each model.

4.2 Log Predictive Density Scores

We compute the average log predictive score over all variables at each time horizon h from $1, \dots, 20$, as well as the overall average (over the horizons) for each model in Table 4.3. The TVP-SV-VAR-NG outperforms all models, as it has the highest overall LPDS, meaning that density forecasts from the normal-gamma are superior to the remaining models, on average. Runner-up models include TVP-SV-VAR-TG and the CC-FSV-VAR-HS, with the triple-gamma maintaining a significant edge.

The relatively high LPDS for the TVP-SV-VAR-BL, TVP-SV-VAR-HS, and TVP-SV-VAR-TG is starkly contrasted by their extremely poor point-forecasting performance. Such a discrepancy could be attributed to our use of an approximation to the posterior predictive distribution in equation 2.4, instead of a more exact approach. Alternatively, we may interpret this result directly, and conclude that TVP-SV-VAR models have more accurate density forecasts due to their shrinkage properties, despite their point-forecasting underperformance.

All models include a large decrease in the LPDS at horizons 5 and 6. These dates correspond with 2020Q2 and 2020Q3, an unprecedented time in economic and financial markets. During this time period, all macro-economic variables experienced huge negative shocks due to the COVID-19 pandemic. These were quite volatile times, as unemployment rates quickly increased, the Federal Reserve enacted significant monetary policy expansion and the price of crude oil became negative. Given the unprecedented and global nature of these shocks, our models' posterior distributions were not equipped to accurately predict this behavior. The superior performance of the TVP-SV-VAR-NG is quite impressive, given the economic setting.

4.3 Factor Loadings

To better understand the factor stochastic volatility, we examine the posterior median of the factor loadings, columns of Λ in equation 1.9. The factor loadings provide an interpretable understanding of the phenomena being captured by our factors. We examine the factor loadings of the VAR-FSV-HS as this model seems to accurately generate point forecasts, according to both RMSFE measures. We standardize the columns for better interpretability and present their heatmap in figure 4.1. Because these are factor loadings for the time-varying covariance matrix, their values correspond to the changing size of the residual errors, and not the data.

Our focus moves to factors affecting socio-demographic data. Factor 1 appears to be cap-

LPDS	Model No.									
h	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10
1	-1.92	-6.62	11.82	-1.45	12.12	8.65	-0.88	-2.80	0.03	-2.47
2	-2.14	-3.14	2.68	-2.2	5.58	2.42	-12.85	-19.07	-0.6	-14.41
3	-1.86	6.35	1.44	-1.66	4.68	6.15	-26.41	-25.97	-1.19	-15.2
4	-63.83	-82.39	-77.3	-63.74	-70.51	-82.19	-41.37	-35.68	-1.9	-18.99
5	-1408.55	-1648.2	-1620.37	-1399.2	-1292.53	-1557.13	-56.39	-46.98	-2.3	-29
6	-549.32	-595.39	-553.13	-550.86	-488.41	-540.77	-70.74	-58.65	-2.44	-31.44
7	-57.67	-59.7	-45.42	-57.71	-42.31	-69.26	-87.1	-70.33	-2.75	-32.66
8	-57.2	-57.64	-49.85	-57.03	-53.1	-67.84	-104.04	-82.44	-3.34	-43.99
9	-79.43	-74.5	-78.2	-79.64	-74.67	-81.57	-121.17	-94.90	-3.82	-46.16
10	-38.22	-34	-34	-38.05	-26.69	-37.16	-137.96	-107.36	-4.3	-53.91
11	-29.07	-32.33	-27.87	-29.21	-23.28	-31.09	-154.96	-120.12	-4.83	-59.25
12	-59.03	-47.69	-64.12	-59.22	-46.77	-47.25	-172.01	-132.91	-5.43	-66.18
13	-66.82	-49.35	-74.91	-67.78	-56.92	-47.65	-188.84	-145.40	-6.18	-68.38
14	-65.86	-40.61	-62.5	-65.15	-52.64	-38.85	-206.01	-157.59	-6.86	-78.39
15	-51.45	-33.53	-51.43	-51.63	-45.78	-30.16	-222.87	-169.86	-7.6	-80.63
16	-56.12	-39.54	-55.87	-55.94	-47.37	-35.18	-239.92	-181.97	-8.44	-90.38
17	-60.22	-25.7	-45.2	-61.4	-40	-23.13	-256.2	-194.35	-9.16	-92.89
18	-39.94	-13.22	-24.74	-40.4	-24.53	-14.97	-272.81	-207.10	-10.21	-102.65
19	-51.95	-20.72	-35.56	-51.49	-31.55	-18.99	-289.67	-220.61	-11.46	-105.17
20	-43.18	-12.3	-25.73	-42.32	-21.09	-12.26	-306.94	-234.12	-12.35	-115.35
Avg.	-139.19	-143.51	-145.51	-138.81	-120.79	-135.91	-148.46	-115.41	-5.26	-57.38

Table 4.3: Table showing average log predictive density score (LPDS) for all fitted models, for each h step ahead forecast from the model, as well as the average over all horizons. Model numbers correspond to: (1) CC-SV-VAR, (2) CC-SV-VAR-NG, (3) CC-SV-VAR-HS, (4) CC-SV-VAR-SSVS, (5) CC-FSV-VAR-HS, (6) CC-FSV-VAR-NG, (7) TVP-SV-VAR-BL, (8) TVP-SV-VAR-HS, (9) TVP-SV-VAR-NG, (10) TVP-SV-VAR-TG.

turing an inverse relationship between the unemployment rate data and corporate bond credit spreads and the inflation expectations index. As the volatility in corporate bond spreads and inflation expectations increase, this factor is capturing lower uncertainty in the labor market. This is somewhat unusual as one might expect that increasing uncertainty in the cost of borrowing for large corporations should imply an equivalent reluctance to hire additional employees, driving further volatility in the unemployment rates. Conversely, this interpretation would also suggest that as borrowing costs become more certain, businesses are more likely to hire additional employees, leading to larger residual variances in the unemployment rate. This interpretation is more reasonable, and since these factors are unique up to a rotation and sign change (Hosszejni and Kastner, 2021), mathematically sound.

Factors 2 and 3 both capture a positive relationship between the volatility in the share of wealth of bottom 50% of Americans and other macroeconomic variables, including the Federal Funds rate. A positive relationship suggests that the Federal Reserve is more sensitive to those who share the bottom 50% of wealth than those who belong to the top 1%, rapidly changing the Federal Funds rate to accommodate similarly large changes in this demographic’s wealth. Complementary to factors 2 and 3, only factors 9 and 10 significantly capture shocks to the United States’ wealthiest. From these factors we can derive that volatility in the wealth of the 1% is highly correlated to the price of oil and the 10-2 Year treasury credit spread. Surprisingly, there is no evidence suggesting that shocks in either the Federal Funds rate, or the NASDAQ index, have an effect top 1%’s wealth volatility.

Factor 10 indicates a negative relationship between the size of shocks to the wealth of the top 1% and both inflation expectations and the M2 money supply. When the money supply stabilizes, this demographic experiences larger wealth volatility, suggesting they may be “siphoning” money from the broader economy. Conversely, rapid changes in the money supply seem to stabilize the wealth of the top 1%, demonstrating that monetary policy impacts are felt elsewhere. This aligns with our interpretation of factors 2 and 3, where we discuss central

Standardized Posterior Median of Factor Loadings For VAR-FSV-HS

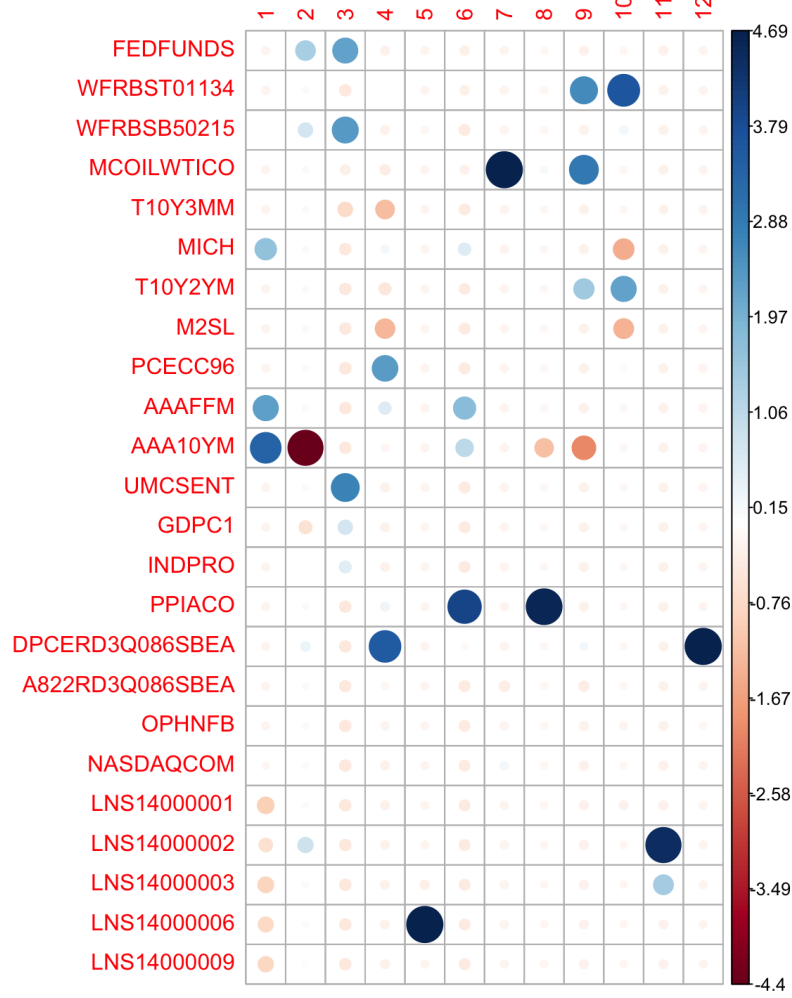


Figure 4.1: A heatmap of the posterior median of factor loadings for the time-varying covariance matrix of residuals for a CC-VAR-FSV-HS. Each column is standardized for interpretability.

bank sensitivity to the wealth of the bottom 50%.

Factor 5 solely captures shocks to African American Unemployment, while Factor 11 jointly (and exclusively) captures shocks to both Women’s Unemployment and White Unemployment. Although we hoped to use the factor approach to isolate relationships across multi-collinear data, these loadings suggest that our goal was not achieved. In principle, we might expect the volatility of all unemployment variables to be generated by a single factor, but fail to observe this outcome. Thankfully, the lack of this behavior allows for interpretation regarding unemployment and monetary policy. The small loading for Women’s Unemployment in Factor 2, shows the existence of a slight correlation to shocks in both the bottom 50%’s share of wealth and the Federal Funds rate. No other factor seems to capture similar behavior, so this should be noteworthy, as it indicates that monetary policy is more sensitive to women’s unemployment rate, as opposed to the overall unemployment rate implied by the Federal Reserve’s dual mandate.

4.4 Impulse Response Functions

When examining impulse response functions, we must be careful to avoid suggesting any causal relationships. Although it might be tempting, as impulse response functions are the response of variables j given a shock in i , central banks have very few policy levers that can be changed to affect the state of the economy. This is why we place so much importance on analyzing the Federal Funds rate, as this is the central bank's main tool for controlling the economy, whose causality should be more straightforward to interpret. We present structural impulse responses from shocks to the Federal Funds rate on the Wealth and Unemployment Data for the VAR-SV-HS and the TVP-SV-VAR-NG models (figure 4.2).

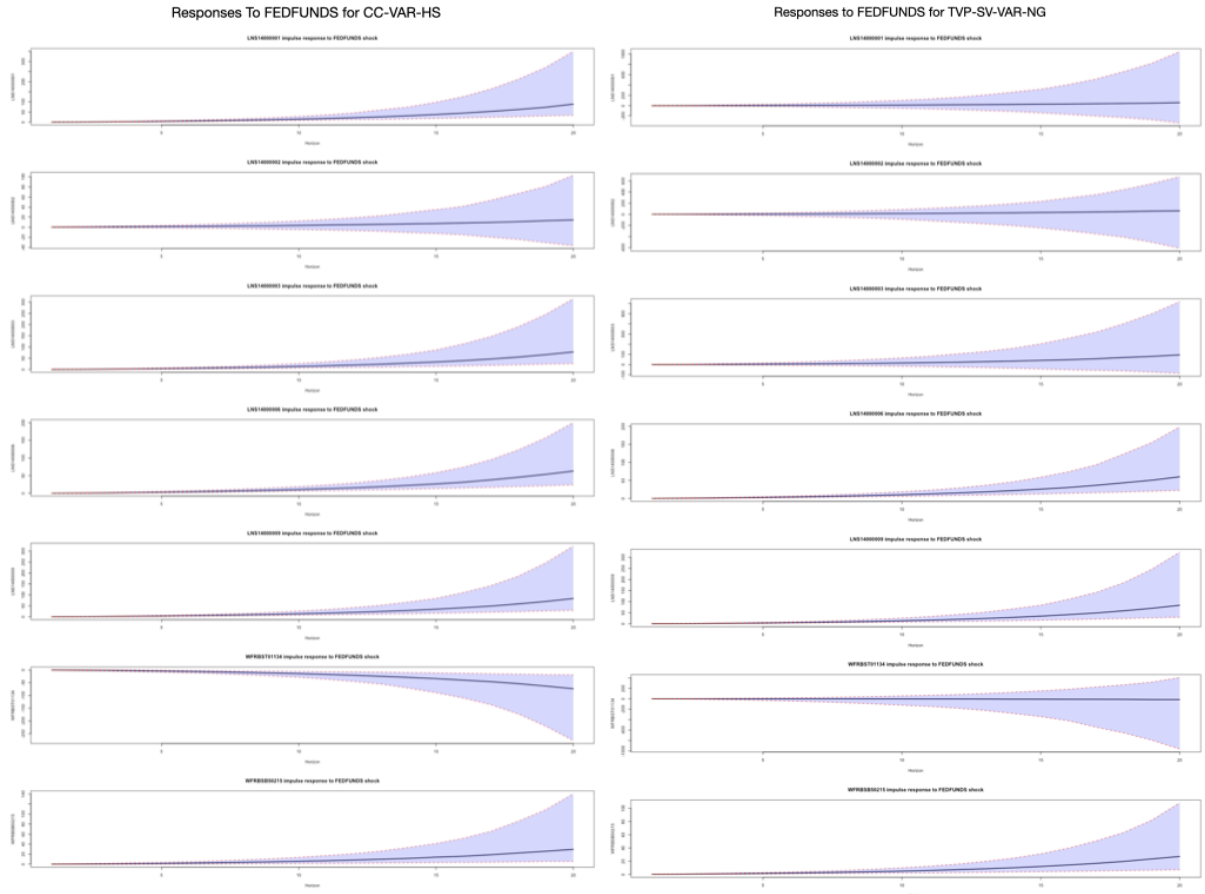


Figure 4.2: Structural Impulse Responses for macro-economic variables of interest, given a 1 standard deviation shock to the Federal Funds Rate for the CC-SV-VAR-HS (left) and the TVP-SV-VAR-NG (right). From top to bottom as: (1) Unemployment Rate - Men (2) Unemployment Rate - Women, (3) Unemployment Rate - White, (4) Unemployment Rate - Black or African American, (5) Unemployment Rate - Hispanic or Latino, (6) Share of Total Net Worth Held by the Top 1 % and (7) Share of Total Net Worth Held by the Bottom 50%.

Unfortunately, these structural impulse responses do not display interesting behavior in the median, and are mostly flat. Regardless, we can make use of the $[0.16, 0.84]$ credible intervals to examine what these impulse response functions may capture. In either model, shocks to the Federal Funds rate appear to negatively impact the wealth of the top 1%. Although the median effect is small, the credible interval bands indicate that there are fatter tails favoring a negative response. On the other hand, the skew of credible interval bands for the wealth of the bottom 50% favors a positive response, similar to the interpretation of the factor loadings.

In the median, unemployment data also exhibits negligible change. The impulse response credible interval for women’s unemployment remains uninformative to a shock in the Federal Funds rate, with a symmetric credible band and low median response. This comes in contrast with the evidence of Federal Reserve sensitivity to women’s unemployment discussed for the factor loadings.

The remaining unemployment variables indicate that there is a positive response to a shock in the Federal Funds rate, except for men’s unemployment, with the short tail allowing for negative values. Positive responses are reasonable in light of economic theory and Federal Reserve policy, which might raise the interest rate to fight inflation, causing an increase in the unemployment rate due to a tighter economic environment.

Despite our regularization, impulse responses display no evidence of divergent behavior across our demographics. Details on how to access IRFs for other models is available in Appendix E, and also do not reveal meaningful behavior.

4.5 Structural Matrix for the TVP-SV-VAR-NG

The interpretation of the elements of the structural matrix A_t^{-1} is analogous to that of the factor loadings discussed in section 4.3. The structural matrix directly describes the relationship of the innovation variances amongst our variables. In lieu of the impulse response functions, we can examine the posterior median (over time) of the structural elements of the TVP-SV-VAR-NG, to better understand the structural relationship between the fitted residuals.

Our wealth variables appear to have negligible dependence on the Federal Funds rate, although the top 1%’s errors appear to be marginally negatively impact. Volatility of the bottom 50%’s wealth is correlated with variance in the consumer sentiment index, likely due to representing a larger percentage of the population.

Our focus moves to the unemployment variables. Because elements of A_t^{-1} are computed as time-varying coefficients of the TVP regression with shrinkage, we additionally benefit from the fact that they are regularized. Otherwise, it is likely that these terms may have been overfitted, and lost meaningful interpretability. Thankfully, the normal-gamma regularization has heavily shrunk many of the correlations observed in figure 3.1.

Amongst the unemployment data, few structural relationships remain, aside from Men’s unemployment, White unemployment and Hispanic unemployment, which can be observed by examining the “LNS14000001” column in figure 4.3. While the existence of this relationship is to be expected, given that a proportion of men in the United States will inevitably be White or Hispanic, the nature of this relationship is noteworthy. It is unusual that the structural coefficient on White unemployment is positive, and simultaneously negative for Hispanic unemployment. This dichotomy implies that increasing variance in Men’s Unemployment increases the volatility of the White unemployment rate, while decreasing the volatility of Hispanic unemployment, a somewhat unusual and unexpected result, given that these population segments overlap.

One possibility is that these structural coefficients are capturing omitted variable bias, as the White and Hispanic unemployment rates were not further disaggregated by gender due to model size constraints and lack of sufficient observations. This hypothesis is further supported by the lack of non-diagonal structural elements for women’s unemployment, African American unemployment, or White unemployment. If the model was capturing a true relationship between the noise in White or Hispanic unemployment rates, we would expect similar behavior in the residuals of other unemployment variables, namely Women’s unemployment, since it should also contain overlapping data. Unemployed Hispanic Women, for example, contribute to both the

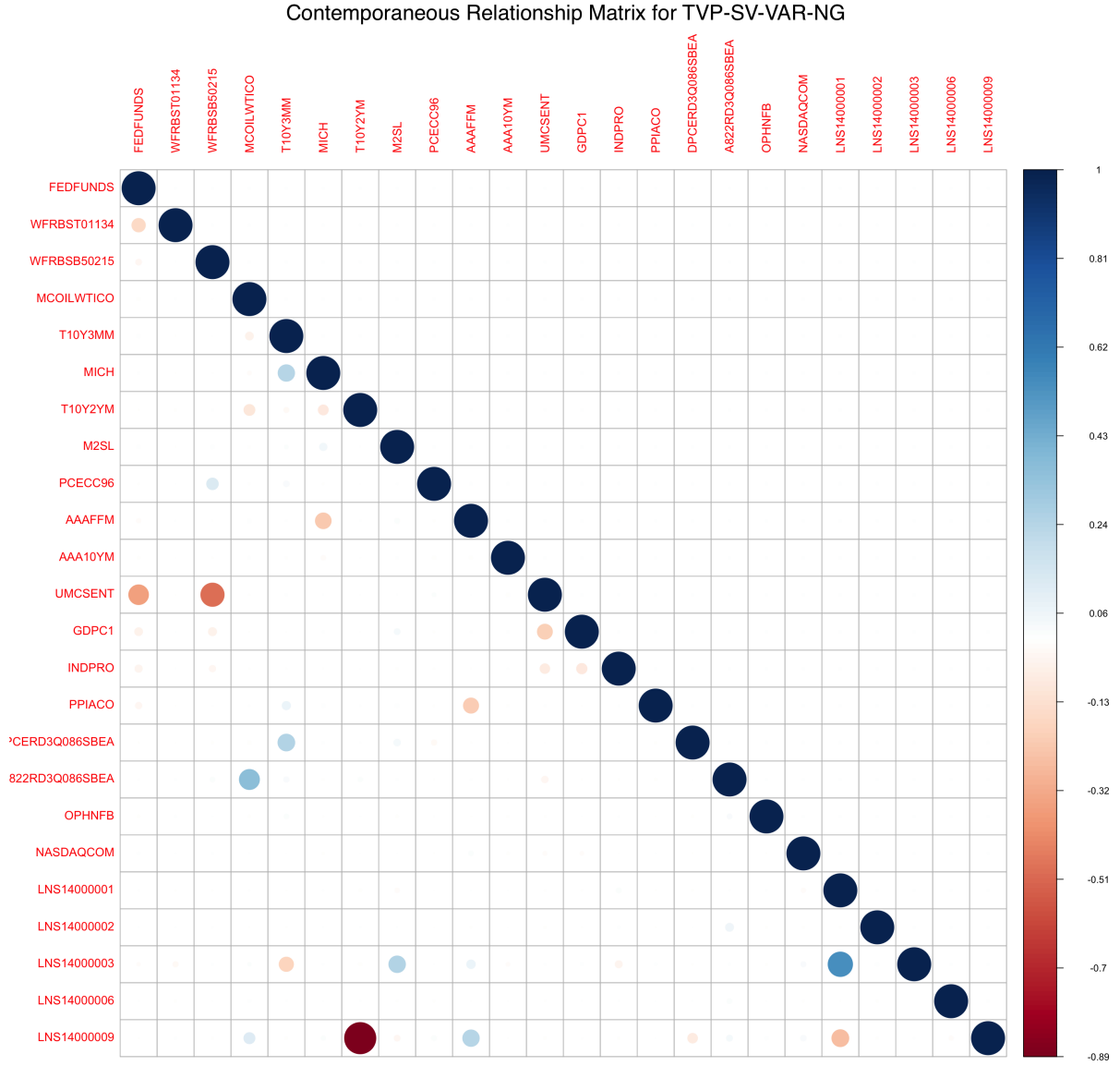


Figure 4.3: Heatmap of the posterior median of the structural matrix A^{-1} , derived from the cholesky decomposition of the covariance matrix of residuals for the TVP-SV-VAR-NG model.

Hispanic and Women’s unemployment rates. Fortunately, this error may be addressed in the future, by including more granular data disaggregated by both gender and socio-demographic groups, such as Hispanic Women’s unemployment rate and White Men’s unemployment rate.

The fact that White, African American, and Women’s unemployment rate appear to only have diagonal structural elements is indicative that the AR(1) process we’ve defined for the log variances is sufficient to capture volatility in these variables, without requiring correlation to other noise terms. This is great news, showcasing the effectiveness of the normal-gamma prior in managing multi-collinearities. Assuming that the issues described above derive from omitted variable bias, we could continue using the normal-gamma prior on a richer dataset to gain a better understanding of the relationship socio-demographic groups have with the macro-economy.

As mentioned several times throughout this paper, the lower triangular structure of the structural matrix implies that there is a sensitivity to the ordering of the variables in our dataset. Under a different variable ordering, we might expect that different coefficients will be

shrunk by the normal-gamma prior, leading to a different interpretation. More importantly, this sensitivity to ordering makes it difficult to determine whether or not the omitted variable bias discussed above is truly present, or just a consequence of improper ordering.

Chapter 5

Conclusions

The aim of this paper was to evaluate the use Bayesian regularization methods in a TVP-SV-VAR model under extreme multi-collinearity, in order to gain interpretable knowledge about the relationship different socio-economic and socio-demographic groups have with the economy. We used popular regularization approaches for constant coefficient and time-varying parameter VAR models, as well as different stochastic volatility specifications, to achieve our goal. We found that, for constant coefficient models, those with horseshoe shrinkage priors performed the best with respect to all three of our forecasting metrics. For TVP-VAR models, normal-gamma shrinkage performed the best whereas the remaining models heavily under-performed.

Using the factor loadings and fitted structural matrix for the aforementioned best-performing models, we were able to recover some results from empirical economic theory regarding the relationship between interest rates, credit spreads, inflation expectations and unemployment. We used these values to explore wealth distribution variables, and found that the share of total net worth of the United States' top 1% is more sensitive to shocks in interest rates and credit spreads, than to quarterly returns of the NASDAQ stock index. We also uncovered an interesting dichotomy between the correlation of errors of different socio-demographic unemployment rates. Through this we discuss the presence of omitted variable bias in our model, and suggest that a larger dataset may amend this issue. On the other hand, impulse response analysis was not insightful for this setting.

Overall, the performance of our time-varying parameter models leaves a lot to be desired. They require a significant number of parameters to define and thus, posterior sampling is quite computationally burdensome on consumer hardware. This also severely restricts the size of the dataset that can be used, and the corresponding conclusions that can be inferred about the data. The poor performance of the triple-gamma prior is quite disappointing, as its flexibility and theoretical posterior distribution properties should have allowed it to provide improved inference and insight by inducing shrinkage in a compelling manner.

Future research could explore the use of dynamic shrinkage processes, such as those in [Kalli and Griffin \(2014\)](#); [Hauzenberger et al. \(2024\)](#); [Knaus and Frühwirth-Schnatter \(2023\)](#), on multi-collinear data, as these approaches allow the amount of shrinkage in the model to vary over time. A dynamic triple-gamma process is already implemented for time-varying parameter regressions in the `shrinkTVP` package, and thus, should be fairly straightforward to explore. Additional research to expand the use of factor stochastic volatility in TVP-VAR models may prove useful due to their relative success in constant coefficient models. Finally, further research on the effects of monetary policy on different socio-economic and demographic groups should address omitted variable bias by avoiding the use of mixed datasets, such as analyzing gender and race together, without including intersectional variables.

Appendix A

Some Conditional Posterior Distributions

We thank for [Korobilis and Shimizu \(2022\)](#) for compiling many of these priors and making them easily available to us. We present these priors for constant coefficient VARs using their notation, but they can be easily extended to TVP-VARs by applying them to the non-centered parameterization. [Huber et al. \(2021\)](#) and [Cadonna et al. \(2020\)](#) were also valuable resources.

Hierarchical Bayesian Lasso Conditional Posterior

The [Park and Casella \(2008\)](#) Laplace (double exponential) prior takes the form

$$\beta | \{\tau_j^2\}_{j=1}^p, \sigma^2 \sim N_p(0, \sigma^2 D) \quad (\text{A.1})$$

$$\tau_j^2 | \lambda^2 \sim \text{Exponential}\left(\frac{\lambda^2}{2}\right), \quad \text{for } j = 1, \dots, p \quad (\text{A.2})$$

$$\lambda^2 \sim \text{Gamma}(r, \delta) \quad (\text{A.3})$$

$$\sigma^2 \sim \frac{1}{\sigma^2} \quad (\text{A.4})$$

where $D = \text{diag}(\tau_1^2, \dots, \tau_p^2)$.

The corresponding conditional posteriors are of the form

$$\beta | \bullet \sim N_p(VX^T y, \sigma^2 V) \quad (\text{A.5})$$

$$\frac{1}{\tau_j^2} | \bullet \sim \text{IG}\left(\sqrt{\frac{\lambda^2 \sigma^2}{\beta_j^2}}, \lambda^2\right), \quad \text{for } j = 1, \dots, p \quad (\text{A.6})$$

$$\lambda^2 | \bullet \sim \text{Gamma}\left(r + p, \frac{\sum_{j=1}^p \tau_j^2}{2} + \delta\right) \quad (\text{A.7})$$

$$\sigma^2 | \bullet \sim \text{Inv-Gamma}\left(\frac{n + p}{2}, \frac{\Psi + \beta' D^{-1} \beta}{2}\right), \quad (\text{B.25})$$

where $V = (X^T X + D^{-1})^{-1}$, $D = \text{diag}(\tau_1^2, \dots, \tau_p^2)$, and $\Psi = (y - X\beta)'(y - X\beta)$.

Normal-Gamma Conditional Posterior

The Normal-Gamma prior of [Brown and Griffin \(2010\)](#) has the form

$$\beta|\{\tau_j^2\}_{j=1}^p \sim N_p(0, D) \quad (\text{A.8})$$

$$\tau|\lambda, \gamma^2 \sim \text{Gamma}\left(\lambda, \frac{1}{2\gamma^2}\right) \quad (\text{A.9})$$

$$\sigma^2 \sim \frac{1}{\sigma^2} \quad (\text{A.10})$$

where $D = \text{diag}(\tau_1^2, \dots, \tau_p^2)$.

The conditional posteriors β and σ^2 are of the following form

$$\beta|\bullet \sim N_p(VX^Ty/\sigma^2, V) \quad (\text{A.11})$$

$$\sigma^2|\bullet \sim \text{Inv-Gamma}\left(\frac{n}{2}, \frac{1}{2}(y - X\beta)'(y - X\beta)\right) \quad (\text{A.12})$$

where $V = (X^TX/\sigma^2 + D^{-1})^{-1}$.

The parameters $\tau_1, \dots, \tau_p$ can be updated in a block. The full conditional distribution of τ_j follows a generalized inverse Gaussian distribution

$$\tau_j|\bullet \sim \text{GIG}\left(\lambda - 0.5, \frac{1}{\gamma^2}, \beta_j^2\right), \quad j = 1, \dots, p \quad (\text{A.13})$$

Horseshoe Conditional Posterior

The horseshoe prior on a regression coefficient β takes the following hierarchical form

$$\beta|\{\lambda_j\}_{j=1}^p, \tau \sim N(0, \sigma^2\tau^2A) \quad (\text{A.14})$$

$$\lambda_j|\tau \sim C^+(0, 1), \quad \text{for } j = 1, \dots, p \quad (\text{A.15})$$

$$\tau \sim C^+(0, 1) \quad (\text{A.16})$$

where $A = \text{diag}(\lambda_1^2, \dots, \lambda_p^2)$, and $C^+(0, \alpha)$ is the half-Cauchy distribution. Thus λ_j has conditional prior density:

$$\lambda_j|\tau = \frac{2}{\pi\tau \left(1 + \left(\frac{\lambda_j}{\tau}\right)^2\right)}. \quad (\text{B.137})$$

In [Makalic and Schmidt \(2016\)](#) and Schmidt (2016) note that the half-Cauchy distribution can be written as a mixture of inverse-Gamma distributions. Specifically:

$$x^2|z \sim \text{Inv-Gamma}(1/2, 1/z), \quad z \sim \text{Inv-Gamma}(1/2, 1/\alpha^2) \quad (\text{A.17})$$

then $x \sim C^+(0, \alpha)$. Therefore, we can formulate the horseshoe prior as

$$\beta|\{\lambda_j\}_{j=1}^p, \tau, \sigma^2 \sim N(0, \sigma^2\tau^2\Lambda), \quad (\text{B.139})$$

$$\lambda_j^2|v_j \sim \text{Inv-Gamma}(1/2, 1/v_j), \quad \text{for } j = 1, \dots, p \quad (\text{A.18})$$

$$v_j \sim \text{Inv-Gamma}(1/2, 1), \quad \text{for } j = 1, \dots, p \quad (\text{A.19})$$

$$\tau^2|\xi \sim \text{Inv-Gamma}(1/2, 1/\xi) \quad (\text{A.20})$$

$$\xi \sim \text{Inv-Gamma}(1/2, 1) \quad (\text{A.21})$$

$$\sigma^2 \sim \frac{1}{\sigma^2} \quad (\text{A.22})$$

where $\Lambda = \text{diag}(\lambda_1^2, \dots, \lambda_p^2)$.

The conditional posteriors are of the form

$$\beta|\bullet \sim N_p(VX^Ty, \sigma^2V) \quad (\text{A.23})$$

$$\lambda_j^2|\bullet \sim \text{Inv-Gamma}\left(1, 1/v_j + \frac{\beta_j^2}{2\tau_j^2\sigma^2}\right), \quad \text{for } j = 1, \dots, p \quad (\text{A.24})$$

$$v_j|\bullet \sim \text{Inv-Gamma}\left(1, 1 + \frac{1}{\lambda_j^2}\right), \quad \text{for } j = 1, \dots, p \quad (\text{A.25})$$

$$\tau^2|\bullet \sim \text{Inv-Gamma}\left(\frac{p+1}{2}, 1/\xi + \frac{1}{2\sigma^2} \sum_{j=1}^p \frac{\beta_j^2}{\lambda_j^2}\right) \quad (\text{A.26})$$

$$\xi|\bullet \sim \text{Inv-Gamma}\left(1, 1 + \frac{1}{\tau^2}\right) \quad (\text{A.27})$$

$$\sigma^2|\bullet \sim \text{Inv-Gamma}\left(\frac{n+p}{2}, \frac{\Psi + \beta'D^{-1}\beta}{2}\right) \quad (\text{A.28})$$

where $V = (X^TX + D^{-1})^{-1}$, $D = \text{diag}(\tau^2\lambda_1^2, \dots, \tau^2\lambda_p^2) = \tau^2\Lambda$, and $\Psi = (y - X\beta)'(y - X\beta)$.

Normal-Gamma-Gamma Conditional Posterior

The normal-gamma-gamma prior takes the form

$$\beta|\{\tau_j^2\}_{j=1}^p, \sigma^2 \sim N_p(0, \sigma^2 D_\tau) \quad (\text{A.29})$$

$$\tau_j^2|\lambda_j \sim \text{Gamma}(a, \lambda_j), \quad \text{for } j = 1, \dots, p \quad (\text{A.30})$$

$$\lambda_j|\varphi \sim \text{Gamma}(b, \varphi), \quad \text{for } j = 1, \dots, p \quad (\text{A.31})$$

$$\varphi \sim \text{Gamma}\left(\frac{1}{2}, \omega\right) \quad (\text{A.32})$$

$$\omega \sim \text{Gamma}\left(\frac{1}{2}, 1\right) \quad (\text{A.33})$$

$$\sigma^2 \sim \frac{1}{\sigma^2} \quad (\text{A.34})$$

where $D_\tau = \text{diag}(\tau_1^2, \dots, \tau_p^2)$.

The conditional posteriors are of the form

$$\beta|\bullet \sim N_p(VX^Ty, \sigma^2V) \quad (\text{A.35})$$

$$\tau_j^2|\bullet \sim \text{GIG}\left(a - \frac{1}{2}, 2\lambda_j, \frac{\beta_j^2}{\sigma^2}\right), \quad \text{for } j = 1, \dots, p \quad (\text{A.36})$$

$$\lambda_j|\bullet \sim \text{Gamma}(a + b, \tau_j^2 + \varphi), \quad \text{for } j = 1, \dots, p \quad (\text{A.37})$$

$$\varphi|\bullet \sim \text{Gamma}\left(pb + \frac{1}{2} \sum_{j=1}^p \lambda_j + \omega\right) \quad (\text{A.38})$$

$$\omega|\bullet \sim \text{Gamma}(1, \varphi + 1) \quad (\text{A.39})$$

$$\sigma^2|\bullet \sim \text{Gamma}\left(\frac{n+p}{2}, \frac{\Psi + \beta'D_\tau^{-1}\beta}{2}\right) \quad (\text{A.40})$$

where $V = (X^TX + D_\tau^{-1})^{-1}$, $D_\tau = \text{diag}(\tau_1^2, \dots, \tau_p^2)$, and $\Psi = (y - X\beta)'(y - X\beta)$.

Stochastic Search Variable Selection Conditional Posterior

Consider the following stochastic search variable selection prior with fixed values of the prior variances.

$$\beta_j | \sigma^2, \gamma_j = 0 \sim N(0, \sigma^2 \tau_{0j}^2) \quad (\text{A.41})$$

$$\beta_j | \sigma^2, \gamma_j = 1 \sim N(0, \sigma^2 \tau_{1j}^2) \quad (\text{A.42})$$

$$P(\gamma_j = 1) = \theta \quad \text{for } j = 1, \dots, p \quad (\text{A.43})$$

$$\theta \sim \text{Beta}(c, d) \quad (\text{A.44})$$

$$\sigma^2 \sim \text{Inv-Gamma}(a, b) \quad (\text{A.45})$$

The non-conjugate prior of [George and McCulloch \(1993\)](#) can be written as

$$\beta | \sigma^2, \gamma, \{\tau_{0j}^2, \tau_{1j}^2\}_{j=1}^p \sim N_p(0, \sigma^2 D), \quad (\text{B.197})$$

where D is a diagonal matrix with diagonal elements $\{(1 - \gamma_j)\tau_{0j}^2 + \gamma_j\tau_{1j}^2\}_{j=1}^p$.

The conditional posteriors are of thus

$$\beta | \bullet \sim N_p(VX^T y, \sigma^2 V), \quad \text{where } V = (X^T X + D^{-1})^{-1} \quad (\text{A.46})$$

$$\sigma^2 | \bullet \sim \text{Inv-Gamma}\left(a + \frac{n}{2} + \frac{p}{2}, b + \frac{1}{2} [(y - X\beta)'(y - X\beta) + \beta' D^{-1} \beta]\right) \quad (\text{A.47})$$

$$\gamma_j | \bullet \sim \text{Bernoulli}\left(\frac{\phi(\beta_j | 0, \sigma^2 \tau_{1j}^2) \theta}{\phi(\beta_j | 0, \sigma^2 \tau_{1j}^2) \theta + \phi(\beta_j | 0, \sigma^2 \tau_{0j}^2) (1 - \theta)}\right), \quad \text{for } j = 1, \dots, p \quad (\text{A.48})$$

$$\theta | \bullet \sim \text{Beta}\left(c + \sum_{j=1}^p \gamma_j, d + \sum_{j=1}^p (1 - \gamma_j)\right), \quad \text{for } j = 1, \dots, p \quad (\text{A.49})$$

where $\phi(x|m, v)$ is the normal density with mean m and variance v .

Appendix B

Additional Information for the Triple-Gamma Prior

Information is taken directly from [Cadonna et al. \(2020\)](#).

Prior for $\sqrt{\theta_j}$		a^ξ	c^ξ	κ_B^2
$\mathcal{N}\left(0, \psi_j^2\right), \psi_j^2 \sim GG\left(a^\xi, c^\xi, \phi^\xi\right)$	normal-gamma-gamma	a^ξ	c^ξ	$\frac{2c^\xi}{\phi^\xi a^\xi}$
$\mathcal{N}\left(0, \tau^2 \psi_j^2\right), \psi_j \sim t_1$	Horseshoe	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{2}{\tau^2}$
$\mathcal{N}\left(0, \tau^2 \xi_j^2\right), \xi_j^2 \sim \mathcal{G}\left(a^\xi, a^\xi\right)$	double gamma	a^ξ	∞	$\frac{2}{\tau^2}$
$\mathcal{N}\left(0, \tau^2 \xi_j^2\right), \xi_j^2 \sim \mathcal{E}(1)$	Lasso	1	∞	$\frac{2}{\tau^2}$

Table B.1: Special cases of the triple-gamma prior for $\sqrt{\theta_j}$. The normal-gamma-gamma is an equivalent formulation.

An outline of the Triple-Gamma Prior MCMC sampling routine for TVP regressions, summarizing Algorithm 1 of [Cadonna et al. \(2020\)](#):

1. Initialization: Define the following parameters for $j = 1, \dots, d$:

$$\tilde{\tau}_j^2 = \frac{\phi^\tau \xi_j^2}{\lambda_j^2}, \quad \tilde{\xi}_j^2 = \frac{\phi^c \xi_j^2}{\kappa_j^2}$$

These parameters are derived from the global (ϕ^τ, ϕ^c) and local (λ_j^2, ξ_j^2) shrinkage components.

2. Posterior Sampling for Regression Coefficients β_j and Shrinkage Parameters: Sample the regression coefficients $\beta_0, \dots, \beta_d$, and local shrinkage parameters λ_j^2, τ_j^2 , from the posterior:

$$p(\beta_0, \dots, \beta_d, \lambda_j^2, \tau_j^2 | y)$$

Using the methods described in [Bitto and Frühwirth-Schnatter \(2019\)](#), Algorithm 1, Steps (a)-(c). These use multivariate gaussians and generalized inverse gaussians to sample the model coefficients.

3. Sampling the Global Shrinkage Parameter a^τ : Sample a^τ using a random walk Metropolis-Hastings (MH) step:

$$z^\tau = \log\left(\frac{a^\tau}{0.5 - a^\tau}\right)$$

Propose $a^{\tau,(s)} = 0.5e^{z^{(s)}/(1+e^{z^{(s)}})}$, and accept with probability:

$$\min \left\{ 1, \frac{q_a(a^\tau, \xi_j^2|y)}{q_a(a^{\tau,(m-1)}, \xi_j^2|y)} \right\}$$

Update:

$$\phi^\tau = \frac{2c^\tau}{\kappa_j^2 a^\tau}$$

4. Sampling Local Shrinkage ξ_j^2 from the Generalized Inverse Gaussian (GIG): Sample $\tilde{\tau}_j^2$ from a generalized inverse Gaussian (GIG) distribution:

$$p(\tilde{\tau}_j^2 | \beta_j^2, \lambda_j^2, \phi^\tau) \sim GIG \left(a^\tau - \frac{1}{2}, \frac{\lambda_j^2}{\phi^\tau}, \frac{\beta_j^2}{\phi^\tau} \right)$$

The density of the GIG distribution (up to a constant of proportionality) is:

$$p(x) \propto x^{a-1} \exp \left(-\frac{b}{2}x - \frac{c}{2}x^{-1} \right)$$

where $a = a^\tau - \frac{1}{2}$, $b = \frac{\lambda_j^2}{\phi^\tau}$, and $c = \frac{\beta_j^2}{\phi^\tau}$.

5. Sampling c^τ via MH using a Student-t Marginal Distribution: Sample c^τ using a random walk MH step, where:

$$z^c = \log \left(\frac{c^c}{0.5 - c^c} \right)$$

Propose $c^{c,(s)} = 0.5e^{z^{(s)}/(1+e^{z^{(s)}})}$, and accept with probability:

$$\min \left\{ 1, \frac{q_c(c^c, \xi_j^2|y)}{q_c(c^{c,(m-1)}, \xi_j^2|y)} \right\}$$

Update:

$$\phi^c = \frac{2c^\tau}{\kappa_j^2 a^\tau}$$

6. Gamma Sampling for Hyperparameters κ_j^2 : Sample κ_j^2 from a gamma distribution:

$$\kappa_j^2 | \lambda_j^2, \tilde{\tau}_j^2, \phi^c \sim \text{Gamma} \left(\frac{1}{2} + c^c, \frac{\theta_j}{2\phi^c \tilde{\xi}_j^2} \right)$$

7. Sampling d_2 for Further Shrinkage Updates: Sample from the distribution of $d_2 | a^\tau, \phi^c, \kappa_j^2$:

$$\kappa_j^2 | \beta_j^2, \xi_j^2, \phi^\tau, a^\tau \sim G \left(d_2 + \frac{a^\tau}{2\xi_j^2} \right)$$

Update:

$$\phi^\tau = \frac{2c^\tau}{\kappa_j^2 a^\tau}$$

Appendix C

Code for Computing RMSFE and Joint RMSFE For Time-Varying Parameter Models

```
compute_LPDS_from_forecast <- function(actual, all_forecasts,
                                       old_sd){
  prior_means <- sapply(names(all_forecasts), function(x)
    return(colMeans(all_forecasts[[x]]$y_pred)))
  prior_variances <- sapply(names(all_forecasts), function(x)
    return(mat2var(all_forecasts[[x]]$y_pred)))
  return (approx_func(prior_means, prior_variances, actual))
}

approx_func <- function(mean, var, actual){
  result <- log(2*pi) + log(var) + (((actual-mean)^2)/(var))
  return (-0.5*result)
}

mat2var <- function(mat){
  return(apply(mat, 2, sd)^2)
}

to_plot <- c("a_tau", "a_xi", "beta", "beta_const",
            "beta_mean", "c_tau", "c_xi",
            "consts", "kappa2", "kappa2_B",
            "lambda2", "lambda2_B", "tau2",
            "theta_sr", "xi2")

results <- list()
for(i in 1:4){
  name <- model_names[[i]]
  name_rmse <- paste0(name, "_rmse")
  name_rmsfe <- paste0(name, "_rmsfe")
  name_jrmsfe <- paste0(name, "_jrmsfe")
  load(models_file_names[i])
  print("plotting")
  for(p in to_plot){
    png(filename=paste0(density_path, name,
                        "density-plot-", p, ".png"), width = 4000, height = 3000)
```

```

    density_plotter(temp_var, to_plot=p)
    dev.off()
}
png(filename=paste0(state_path, name,
    "state_plot", ".png"), width=4000, height=3000)
state_plotter(temp_var, Sigma=FALSE)
dev.off()
png(filename=paste0(state_path, name,
    "state_plot_Sigma", ".png"), width=4000, height=3000)
state_plotter(temp_var, Sigma=TRUE)
dev.off()
for(p in to_plot){
    png(filename=paste0(heatmap_path,
        name, "heatmap_plot_", p, ".png"), width = 5000, height = 5000)
    TV_heatmap(temp_var, to_plot=p)
    dev.off()
}
print("forecasting")
fcasts <- forecast_shrinkTVPVAR(temp_var, n.ahead=20)
print("fitting")
rm(temp_var)
results[[paste0(name, "_LPL")] <- compute_LPDS_from_forecast(y.test,
    fcasts, prior_sd, rescale=FALSE)

gc()
fcasts <- lapply(names(fcasts), function(x) return(fcasts[[x]]$y_pred))
fcasts2 <- lapply(1:length(fcasts),
    function(x) return(sweep(fcasts[[x]] * prior_sd[x], 1,
        y.test[, x]*prior_sd[x], "-")))
fcasts2 <- sapply(fcasts2, function(x) return(apply(x, 1, median)))

for(h in horizons){
    results[[paste0(name_jrmsfe, h)]] <- sqrt(mean(fcasts2[1:h, ]^2))
}
for(h in c(1, horizons)){
    results[[paste0(name_rmsfe, h)]] <- sqrt(mean(fcasts2[h, ]^2))
}
}

```


Appendix D

Posterior Median Point Estimates for RMSFE and Joint RMSFE and State Plots

Computed RMSFE and Joint RMSFE using the posterior median for the time-varying parameter models with poor fit. The Bayesian Lasso gets an improvement although others do not.

TVP-SV-VAR-BL	1	2	3	4	8	12	16	20
JRMSFE	-	13.45	13.87	31.16	30.57	29.88	30.68	31.66
RMSFE	13.98	12.89	14.67	57.51	26.35	35.86	14.25	16.99

Table D.1: TVP-SV-VAR-BL

TVP-SV-VAR-HS	1	2	3	4	8	12	16	20
JRMSFE	-	4.13e+33	3.37e+33	2.92e+33	2.10e+33	1.74e+33	1.51e+33	1.35e+33
RMSFE	3.95e+09	5.84e+33	1.04e+13	3.42e+09	1.13e+08	5.51e+07	2.19e+09	2.25e+11

Table D.2: TVP-SV-VAR-HS

TVP-SV-VAR-TG-SV	1	2	3	4	8	12	16	20
JRMSFE	-	6.36e+06	1.61e+11	3.24e+11	2.29e+11	1.89e+11	1.64e+11	1.49e+11
RMSFE	8.26e+04	8.99e+06	2.80e+11	5.84e+11	6.12e+06	2.70e+09	2.60e+10	3.36e+05

Table D.3: TVP-SV-VAR-TG-SV

State variable plots for our fitted TVP-SV-VAR models with shrinkage:

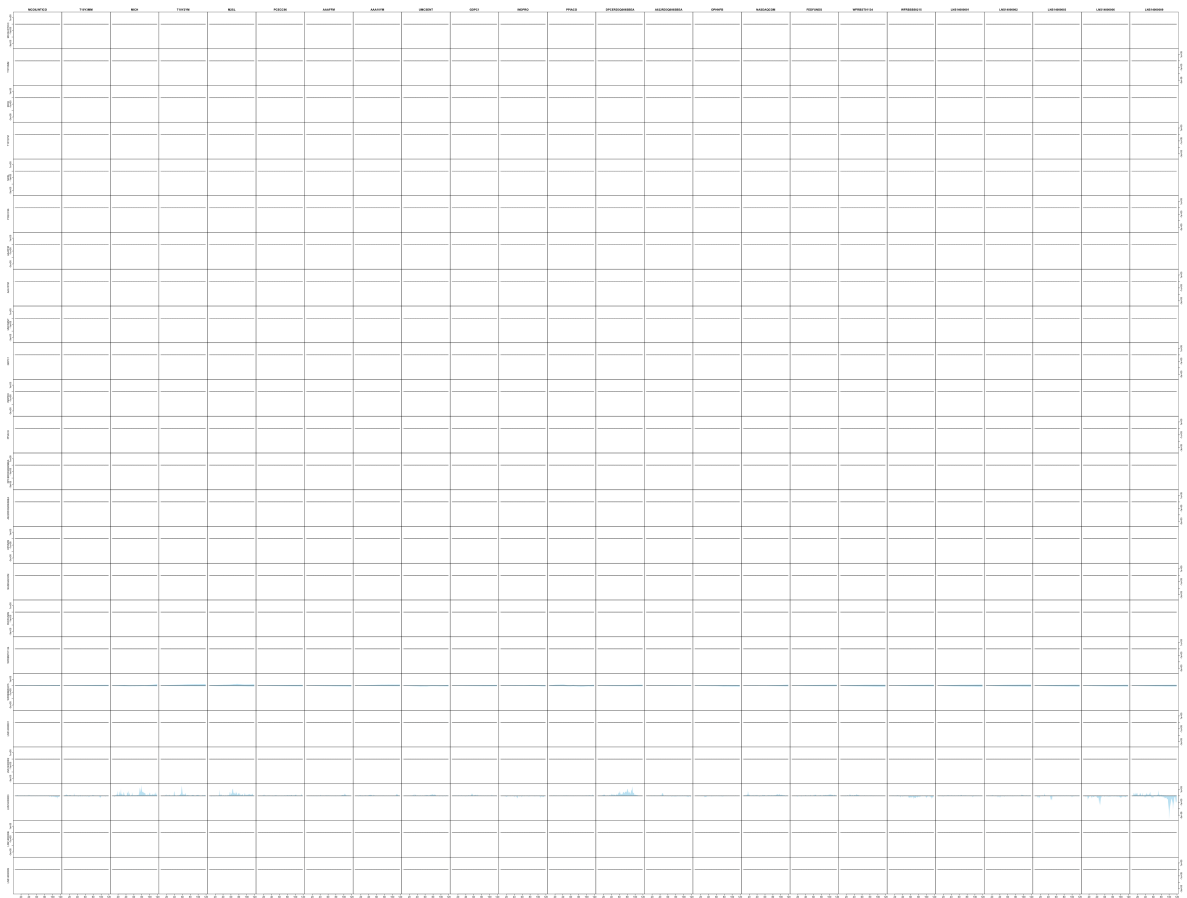


Figure D.1: TVP-SV-VAR-BL State Variable Plot

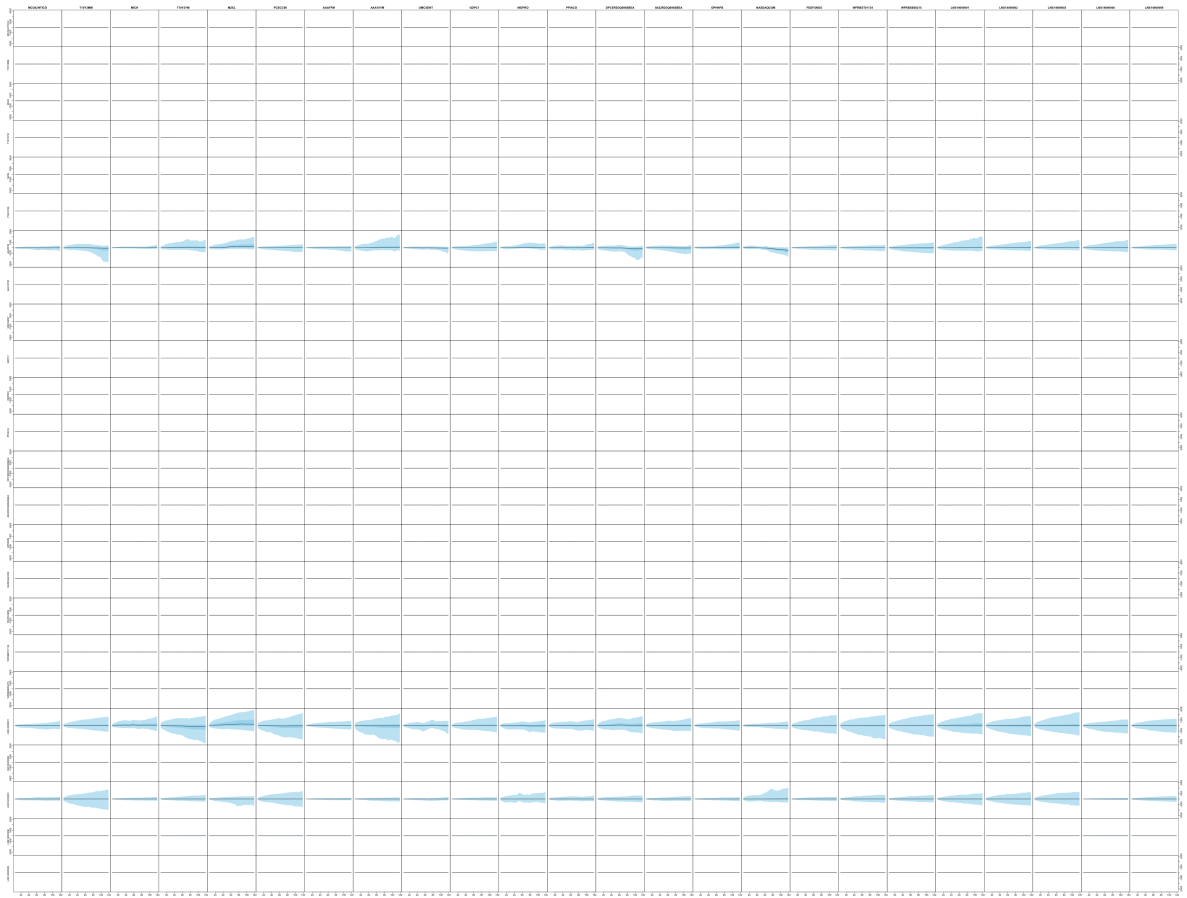


Figure D.2: TVP-SV-VAR-HS State Variable Plot

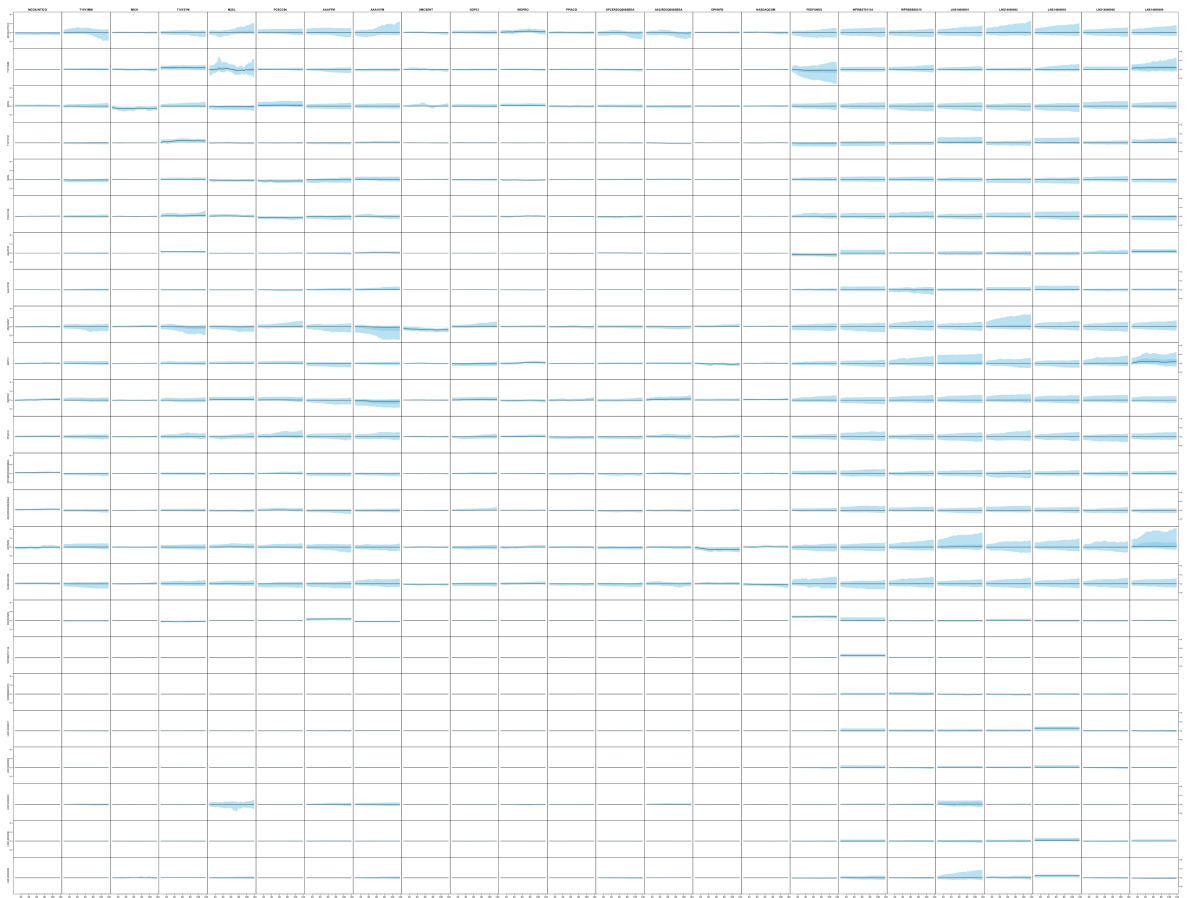


Figure D.3: TVP-SV-VAR-NG State Variable Plot

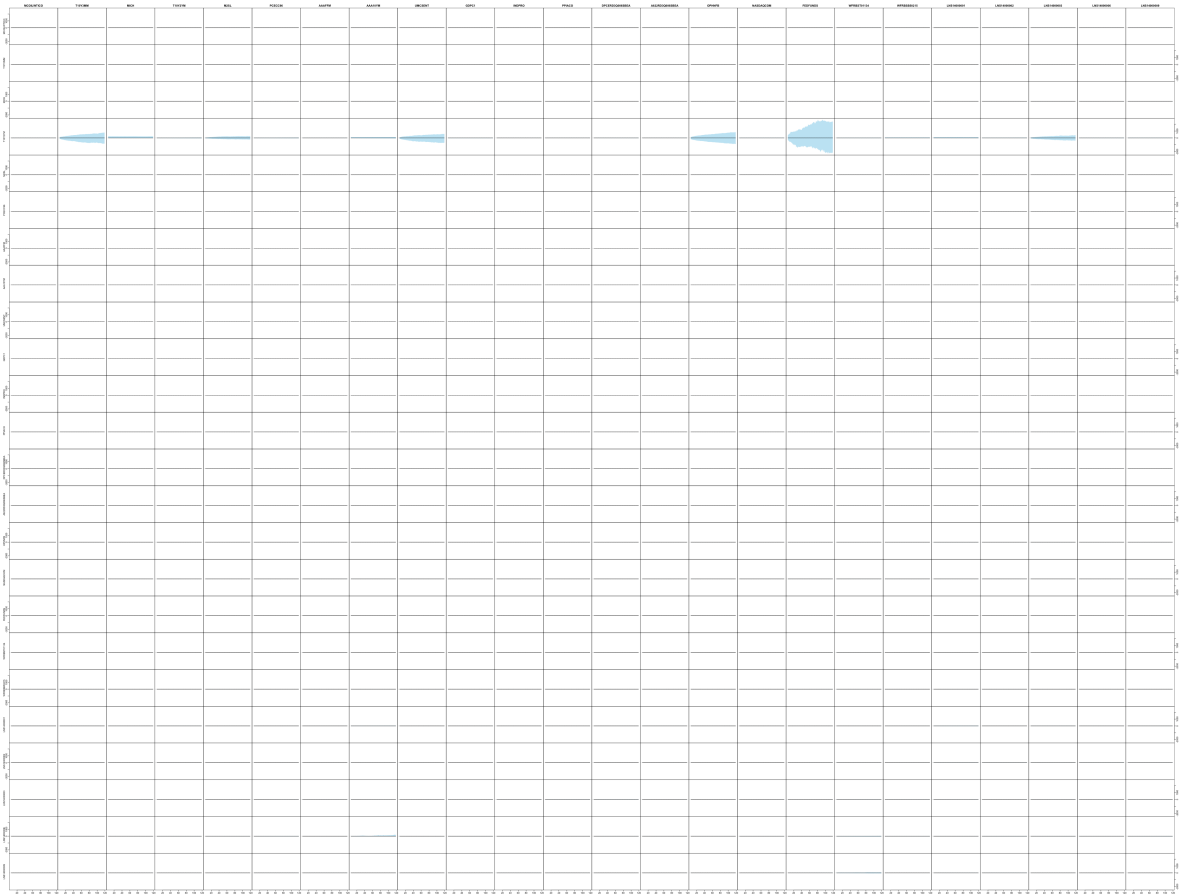


Figure D.4: TVP-SV-VAR-NG State Variable Plot

Appendix E

Supplementary Resources

Please visit the following GitHub repository for plots, code and other files relating to this thesis:
<https://github.com/K2368850/MSc-Thesis>

The last commit as of the submission of this paper: September 15th, 2024

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